



POLITECNICO
MILANO 1863

SCHOOL OF MANAGEMENT
DIPARTIMENTO DI ELETTRONICA,
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 **osservatori.net**
digital innovation



 13.05.22

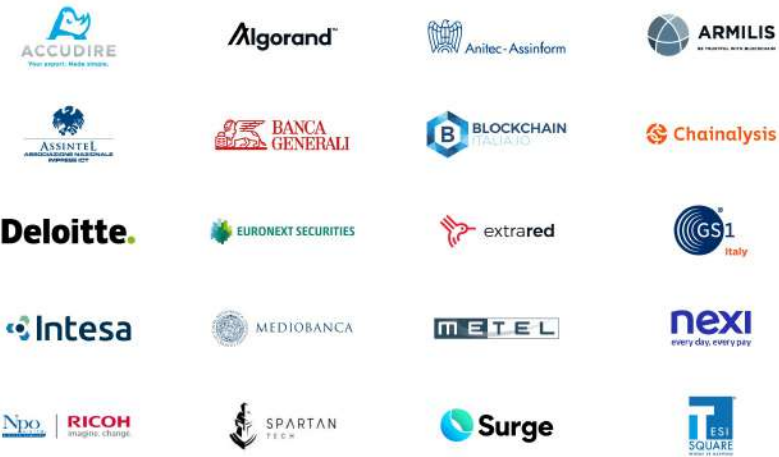
 Blockchain & Distributed Ledger Observatory

DApp & web3: new models and opportunities

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13.05.22

Blockchain & Distributed Ledger Observatory

DApp & web3: new models and opportunities

Workshop starts at 14:00



Set your name and company as your **Zoom** name.
Before speaking, **introduce yourself** with your name and company



To speak write your question **in chat** or **raise your hand** and wait to be called by the organizers



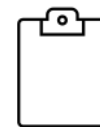
Keep the **audio off** to avoid the echo effect. Activate it only when you take action



If you have any **issues**, try **leaving and re-entering** the meeting



Keep the **webcam disabled**. Activate it only when you take action



Discussions and **polls** will be managed through [slido.com](https://www.slido.com)



Valeria Portale

👤 Director

🔗 Blockchain & Distributed Ledger Obs.

14:00 Opening session

14:10 Welcome and introduction
Valeria Portale, *Blockchain & Distributed Ledger Observatory*

14:20 Web3 and the evolution of Dapp
Giacomo Vella, *Blockchain & Distributed Ledger Observatory*

14:40 Composability in web3: the PoolTogether case
Leighton Cusack, *Co-founder @ PoolTogether*

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Sebastiano Cataudo, *Polygon DAO Core Dev*
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Emmanuel Coen, *Growth @ Hop Protocol*

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Francesco Bruschi, *Blockchain & Distributed Ledger Observatory*
Tommaso Paulon, *Blockchain & Distributed Ledger Observatory*

17:00 End of the day



The **Blockchain Observatory** was founded in 2018 with the mission of **generating and sharing knowledge** on Blockchain technologies. The Observatory **contributes actively to the development of the Italian and international market**, generating contents and events for all the players of the industry

Research

We do **research** to help to understand what Blockchain and DLT technologies are and what opportunities they can generate



Networking

We create **opportunities for cross-industry meetings** and exchanges with Italian and international players, interested in better understanding these issues



Education

We are a **point of reference** for companies that want to keep up-to-date on these issues and learn about market developments



Communication

We create **culture**, communicate and spread knowledge on Blockchain and DLT, raising awareness among companies that could use it



Business

International companies and startups



Use cases around the world



Startups and innovative trends

Internet of Value



Cryptocurrencies



Stablecoins



Cryptoassets



Central Bank Digital Currencies

Blockchain for business



Blockchain ecosystems



Supply Chain

Decentralized web



SSI and Digital Identity



DApps and DeFi



NFT and tokenization



Decentralised metaverse



Governance and DAOs

Italian market



Level of adoption by citizens



Level of adoption by companies



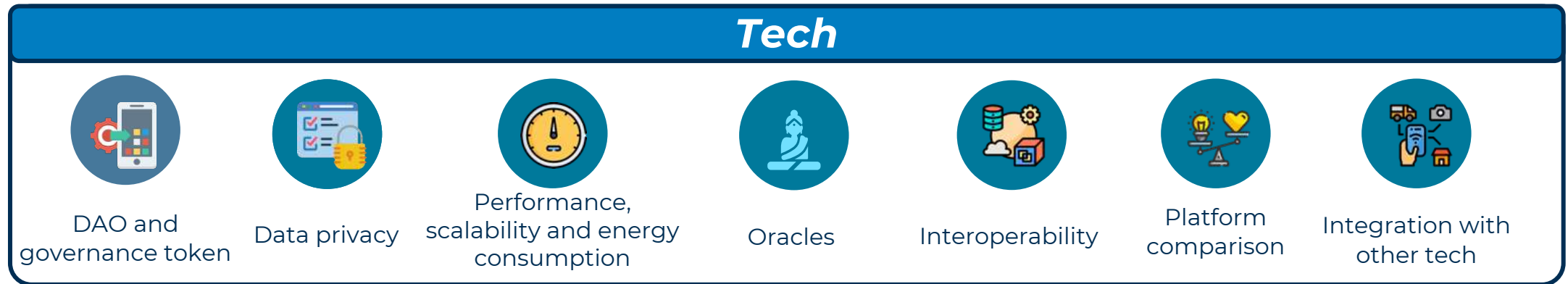
Analysis of the offering



Italian use cases



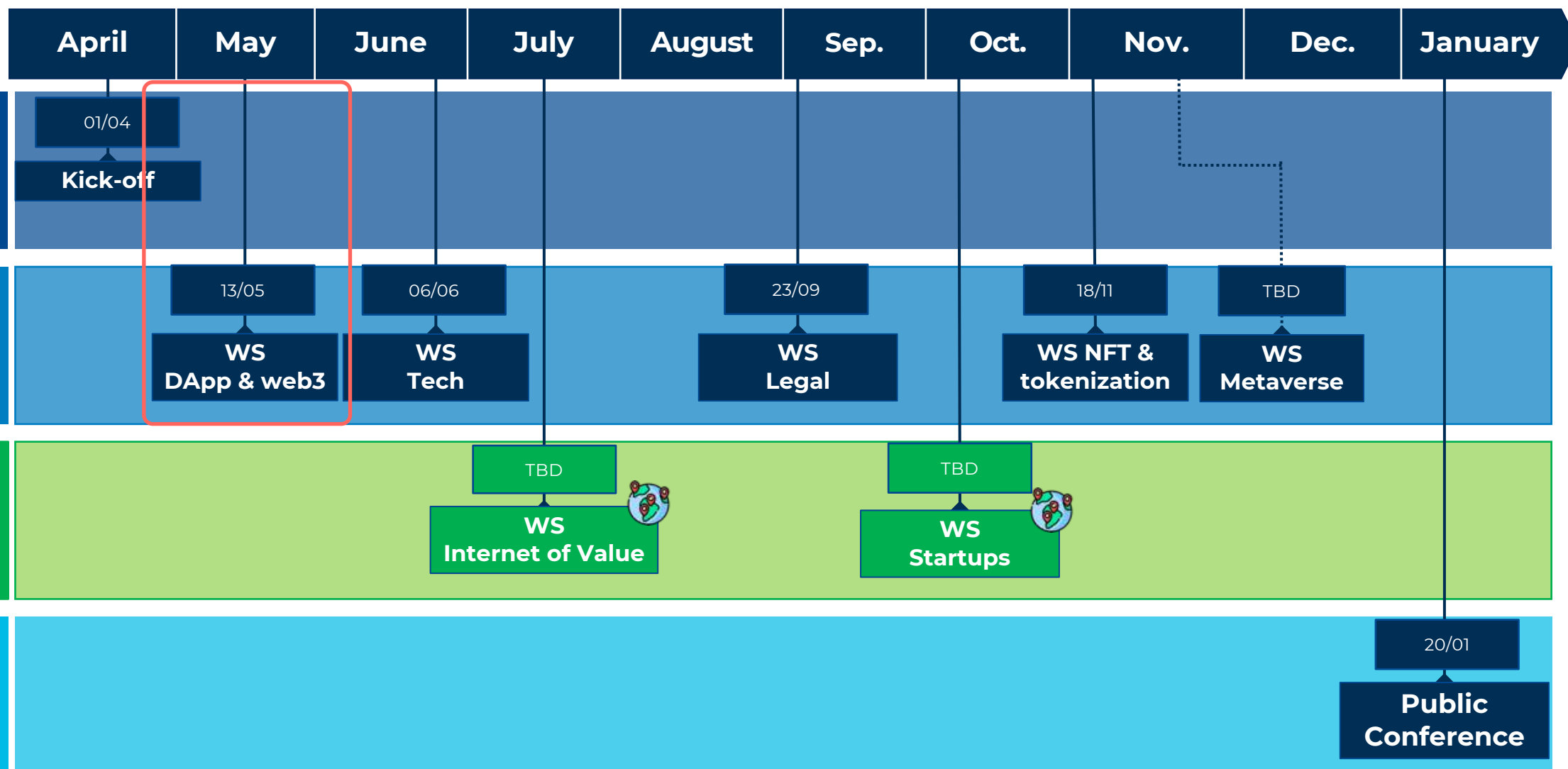
Italian startups



The 2022 activity plan

 Oss. Blockchain & Distributed Ledger

 Workshop DApp & Web3 | 13.05.22





Workshop with
demo e hands on

focus



Goals

- Identifying and testing the infrastructural models of the main DAOs
- Define the advantages introduced by DAOs and any disadvantages compared to traditional organisations
- Define solutions to overcome scalability and privacy issues within a DAO



DAO and governance token

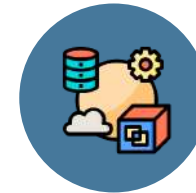
Within the DAO topic, several other areas will be addressed:



Data Privacy



Performance,
scalabilità e consumo
energetico



Interoperabilità



Identity



DApp

Partners and Sponsors

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- **Algorand:** Pietro Grassano
- **Almaviva:** Gianluigi Mascia, Giuseppe Bertone, Geltrude Amodio
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- **NIUMA:** Giampiero Volpi, Pietro Carra
- **NPO Sistemi:** Andrea Colmegna
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- **POMIAGER:** Giuseppe Adduce
- **Poste Italiane:** Emiliano Vernini, Rosario De Chiara
- **PWC:** Gianmarco Guazzo
- **Surge:** Piero Bandini
- **TESISQUARE:** Stefania Vanzetti

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- **Regione Lombardia:** Matteo Santoro

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Web3 and the evolution of Dapp

Giacomo Vella

Blockchain & Distributed Ledger Observatory



«Meta Targeting 'Deep Compatibility' With Blockchain, Web3»

decrypt.co, 23/12/2021

«Visa launches NFT program as it considers the digital art a new form of e-commerce»

techcrunch.com, 30/03/2022



PRADA

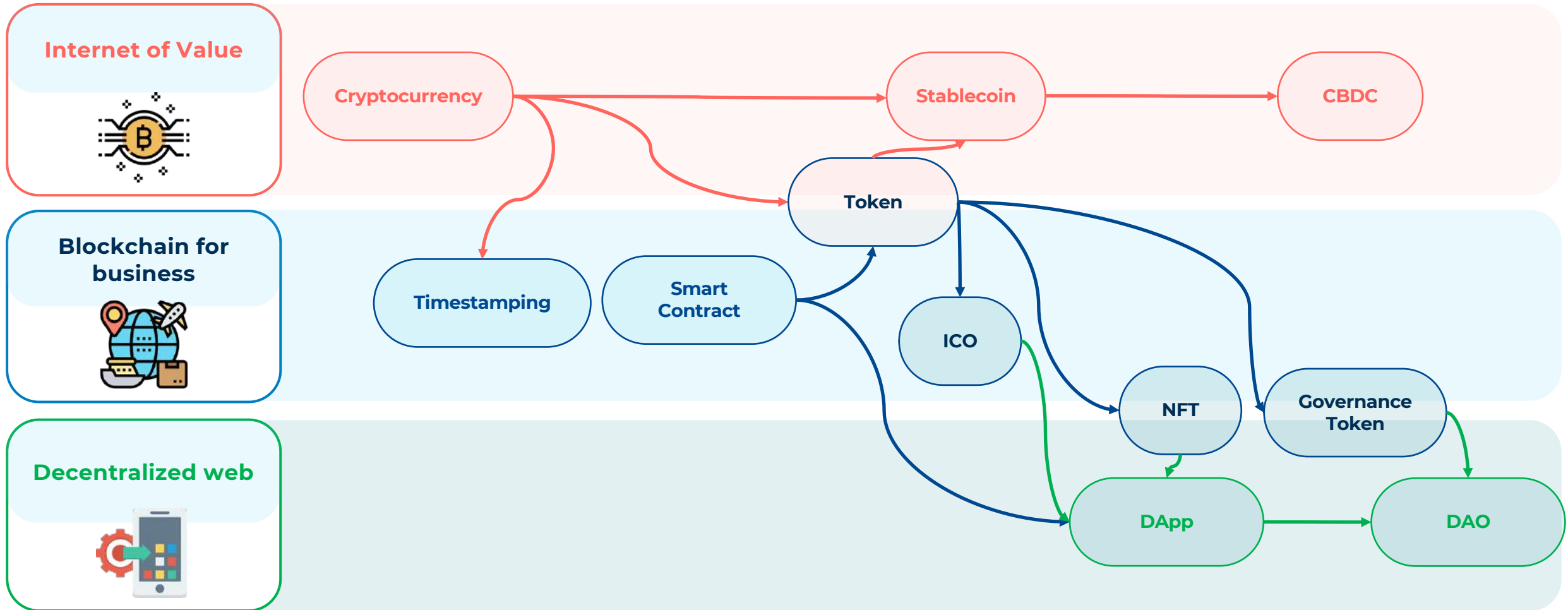
«Adidas and Prada go Web3 with community-focused NFT project»

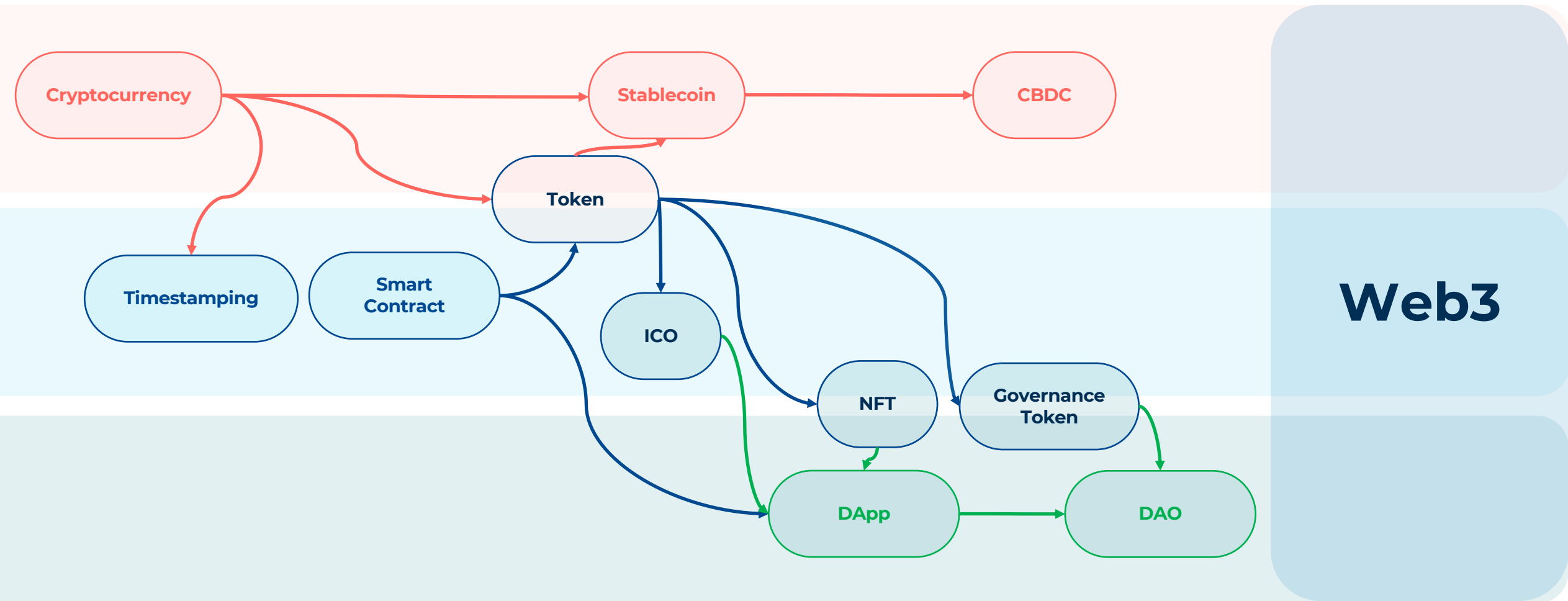
coinrivet.com, 20/01/2022

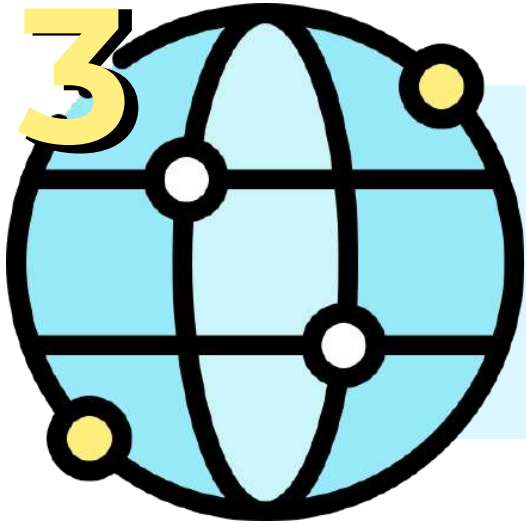
«Switzerland's SEBA Bank wants to use Aave's institutional DeFi platform»

theblockcrypto.com, 21/12/2021



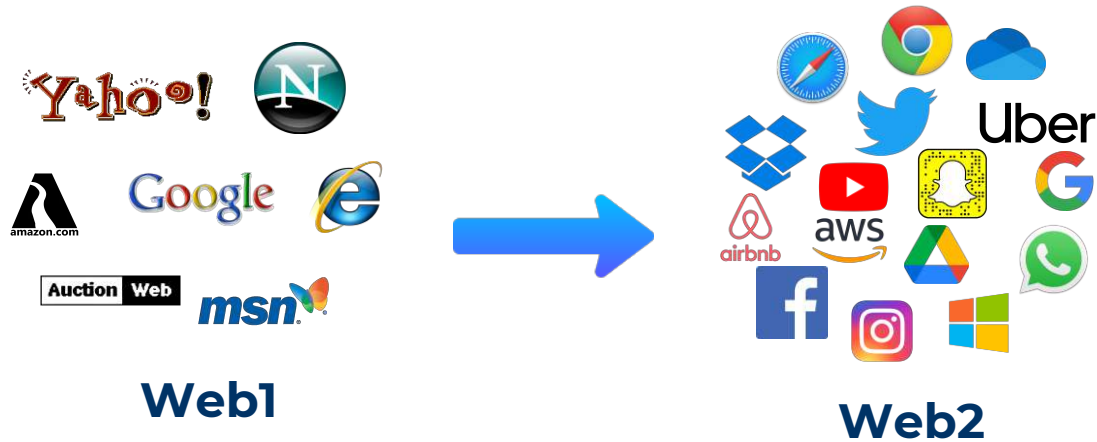






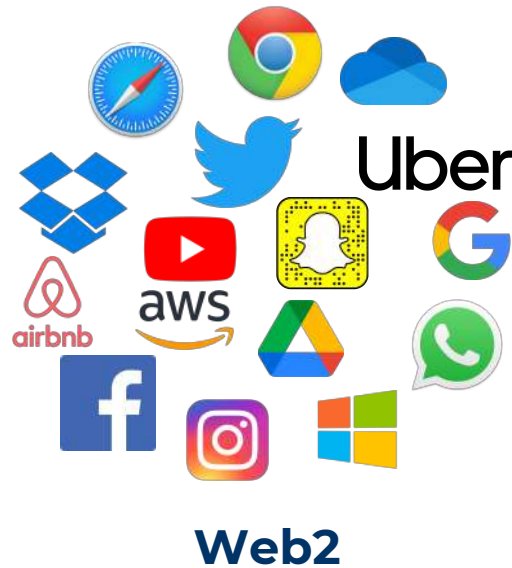
Web3 refers to a potential new iteration of the **internet enabled** by **public blockchains**.

There is still no unique definition of web3

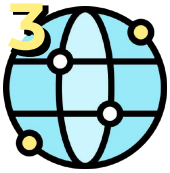


- In **web1** most users could only access information. **Web2** has enabled anyone to **create content** online.
- The term **web2** (or web 2.0) refers to the set of services offered by a **small and defined set of platforms**.
- These **centralised platforms** introduced **major innovations**, such as **social and content aggregation**.





- The **content we create** in the web2 paradigm **does not really belong to us**.
- The user who uses a 'proper' **username on Twitter** may not necessarily also use it on **Facebook** or **YouTube**.
- **Twitter** may **revoke control** of the **name**
- The same thing happens with **identity attributes**.



Web3 enables a transversal user experience across different services, thanks to the possibility of **certifying one's identity** and easily **proving the possession** of certain attributes.



Web1

Read



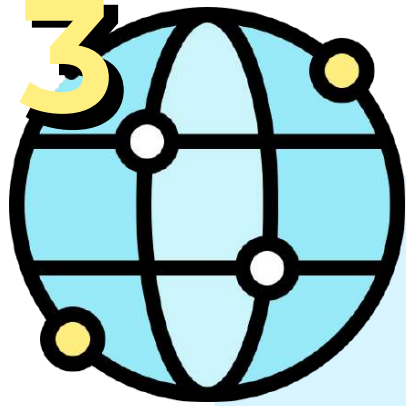
Web2

Read
Write



Web3

Read
Write
Own



The key enablers of web3

- Permissionless
- Smart contracts and Tokens
- A common authentication system (private/public key)
- A native payment system

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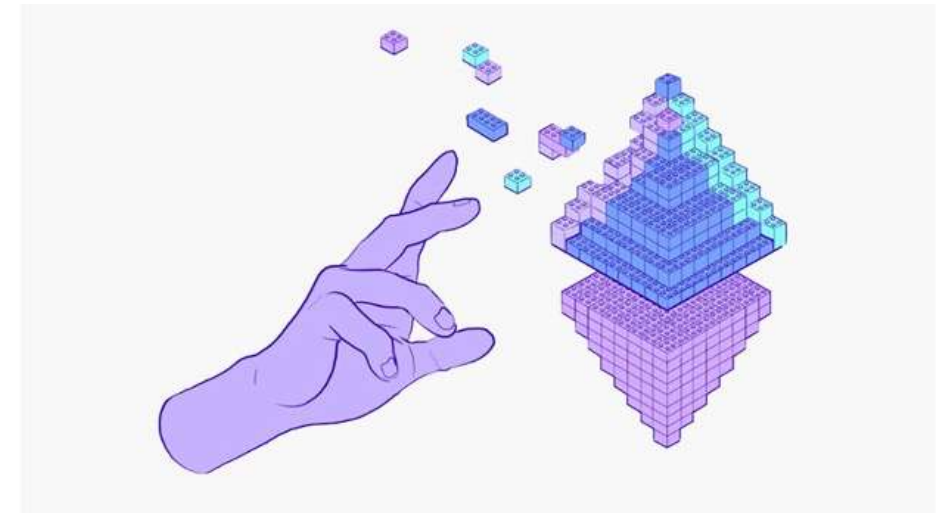


Composability

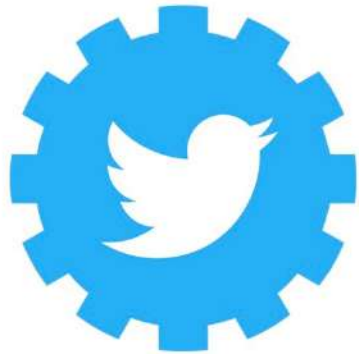
Thanks to the smart contracts' properties, Dapp protocols are **“composable”** and they can be **combined** to create **new instruments and services** that:

- are **functionally interoperable**
- benefit from the **common features** of the **underlying blockchain**

More the protocol is **simple, powerful** and **works well with other protocols**, higher will be considered its **level of composability**.



Source: World Economic Forum, 2021. Decentralized Finance (DeFi) Policy-Maker Toolkit



The concept of **composability** is not new.

Web developers have been building applications from **reusable components** since the **1990s**. In Web 2.0 composable applications were known as "**software mashups**".

Some business started as mashups, but by the start of 2010's **centralized platforms** began to assert their **control on the web**. Some of these companies **limited or even shut down access to their APIs**.

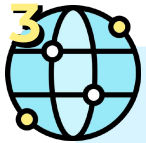
A notable example in the tech industry was **Twitter**, which up till 2011 openly encouraged mashup companies to use its data, but by 2012 was severely curtailing developer access to its API.

Source: <https://thenewstack.io/ceramics-web3-composability-resurrects-web-2-0-mashups/>

An example of working **composability nowadays** is **Stripe**, which provides **payment** support to users throughout a **multitude of web applications**.



However, in **web2**, **composability comes with many limitations**. Traditional applications are inherently permissioned to build upon.



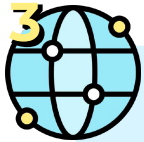
Web3 promises to change this by being **inherently permissionless**, creating an **open foundation** from which any developer can get immediate access to **all protocols** and to an **infrastructure** that operates with **high levels of tamper-resistance and reliability**.

Source: <https://blog.chain.link/defis-permissionless-composability-is-supercharging-innovation/>



Lego blocks have a **rigid framework** to allow any piece to work with the others. Moreover, all **new pieces are always compatible** with existing ones and users can **combine easily** any piece.

This features of **Lego blocks** can be used also to describe **Dapps' protocols** in web3.



In web3 Dapp **developers** can **pick any “block of Lego”** from the ones available in the **blockchain open-source ecosystem**, assemble them and build a **brand-new product**.

This increases the **speed of software innovation** as the work of every developer team is **available to anyone**.

Sources: <https://www.coindesk.com/markets/2019/08/18/the-real-benefits-of-blockchain-are-here-theyre-being-ignored/>
<https://blog.makerdao.com/composable-infrastructure-as-code-an-introduction-to-the-maker-defi-ecosystem/>

Morphological composability

Thanks to the definition of common standards, software components such as **tokens are interoperable** with each other and can **move between** different **decentralized applications**.

Syntactic composability

Ability of Dapps to integrate with each other. **Each smart contract** on the blockchain is public and **can be called by any other**, meaning that any software logic it is available for reuse by the entire ecosystem.

Atomic composability

Several operations across multiple DApps can be **bundled into a single transaction** and **executed together**. If one of the operations fails, then the entire transaction will fail.



Morphological composability: thanks to the definition of common standards, software components such as **tokens are interoperable** with each other and can **move between** different **decentralized applications**.

Increased asset capital efficiency

Maker DAO

A user can gain working capital by **minting decentralized stablecoins** via opening an overcollateralized loan



Aave/Compound

These stablecoins become even more useful when deposited into a **decentralized money market** and tokenized



Other protocols

The newly created **interest bearing tokens**, which represent ownership of the stablecoin deposit, can be **used within other DeFi protocols**

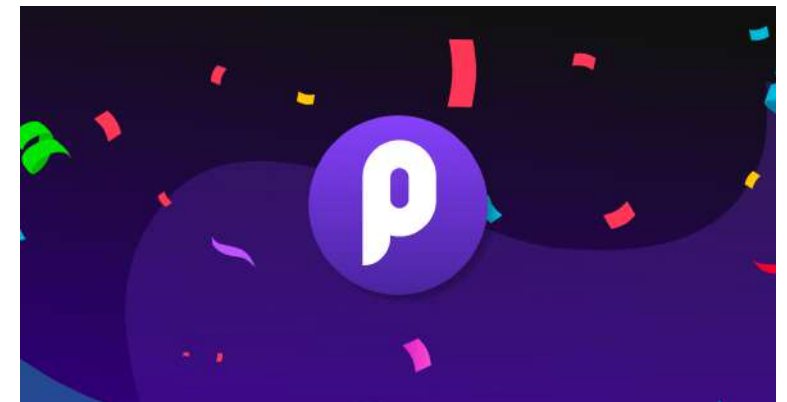


Syntactic composability: ability of Dapps to integrate with each other. **Each smart contract** on the blockchain is public and **can be called by any other**, meaning that any software logic it is available for reuse by the entire ecosystem.

PoolTogether

PoolTogether offer users to earn yield through “no-loss” lotteries. In PoolTogether **users deposit assets** that allow them to take part to the lottery “buying” an entry ticket.

The **protocol takes this deposit and send it to a DeFi protocol** (Compound) where the users’ **asset is deposited and starts to accrue interests**. The total interests will be the lottery pool prize. Then **a tokenized no-loss game ticket is created and sent to the players**. Finally, the **winner is decreed** using Chainlink’s **verifiable random function** (VRF) solution.



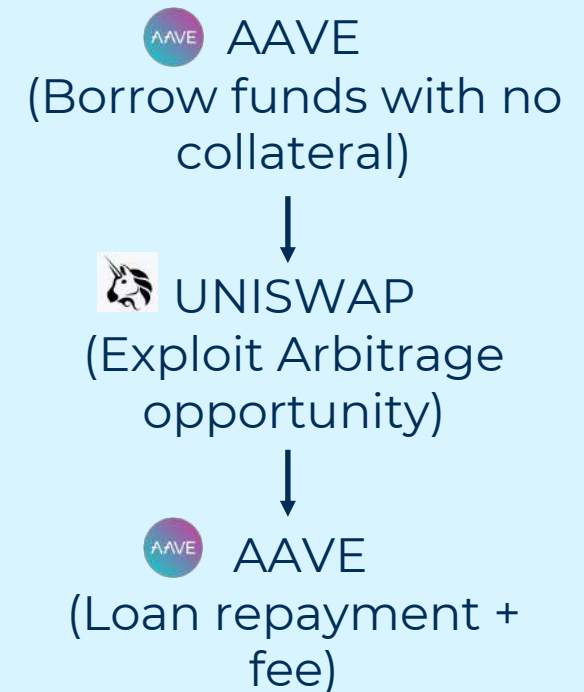


Atomic composability: **everal operations** across multiple DApps can be **bundled into a single transaction** and **executed together**. If one of the operations fails, then the entire transaction will fail.

Flashloans

They are a form of **loan** that allows a user to **borrow funds** and **repay the loan** in the **same transaction**. In this transaction, several operations across multiple Dapps can be **bundled together and executed**. In case the loan is **not repaid** in the same transaction **it will fail**. This allows loans to be **uncollateralized**, meaning that these flash loans are **accessible to anyone** in the world regardless of how much capital they have

Transaction example



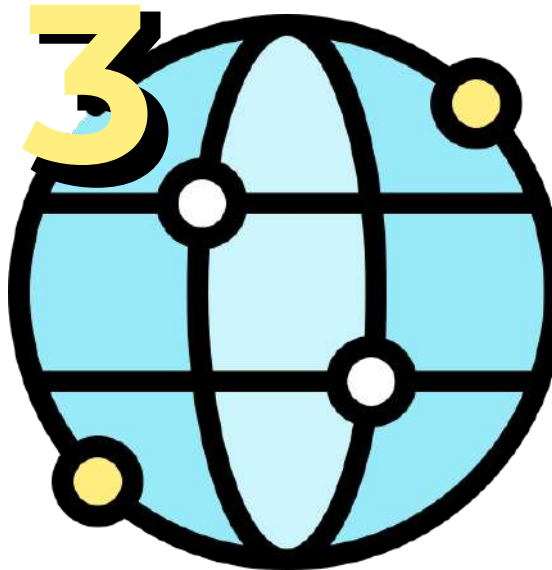
There is still little research on the new architectural, business and governance models enabled by the web3

?

How are companies and startups building new businesses on the web3?

There is still little research on the new architectural, business and governance models enabled by the web3

Permissionless



Common authentication system

Smart contracts and tokens

Native payment system

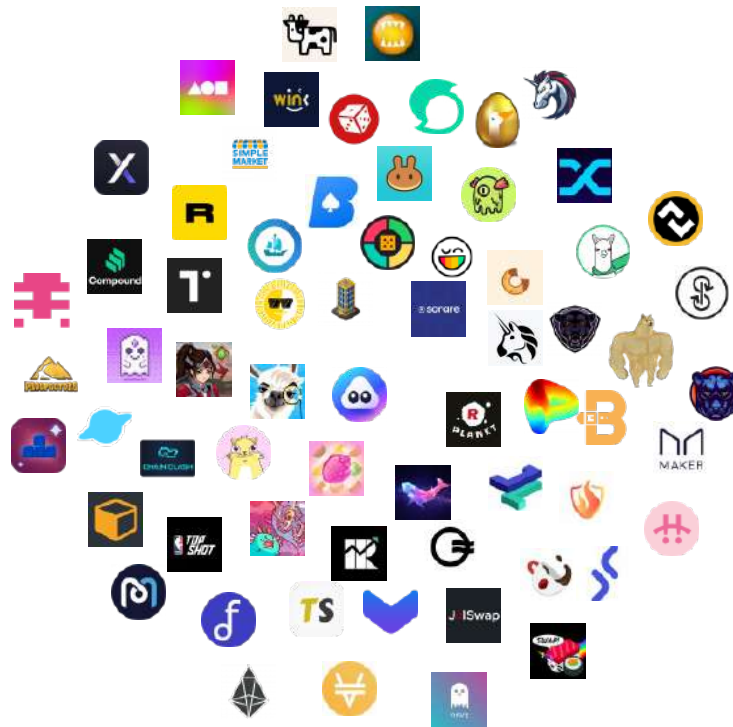
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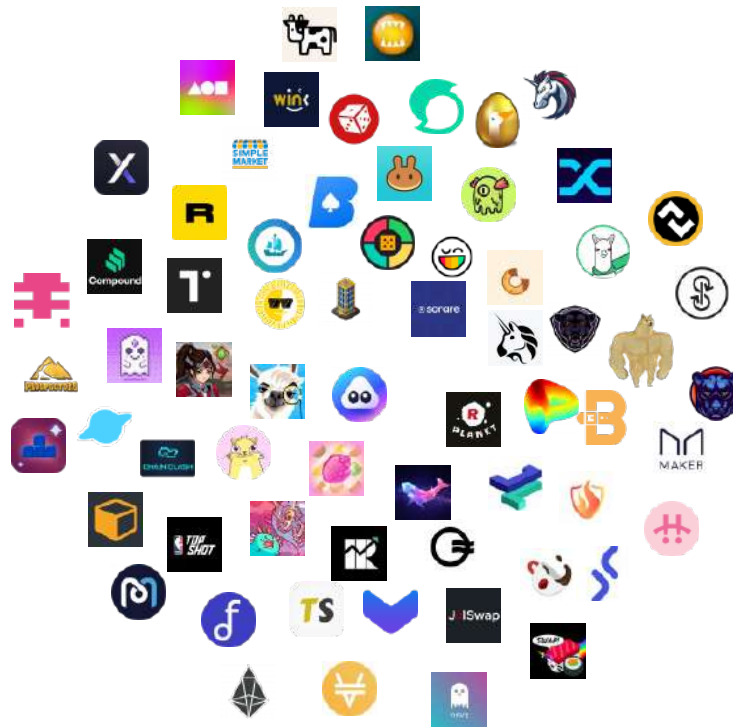


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New revenue models

Decentralized Governance

Increased users' involvement



Use of token

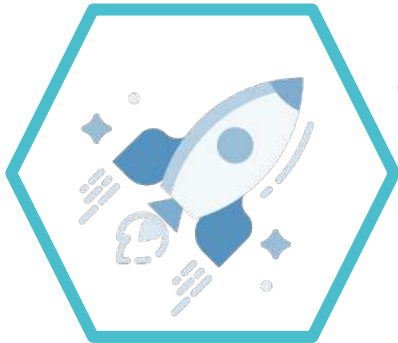
Increased collaboration

New forms of competition



BORDERS

- World
- Main blockchain platforms



CRITERIA

- Top 100 DApps for AMU
- Extraction date 26/04/2022



SOURCE

- DAppRadar

»» **100
DApps** 🔍



DApp websites, white papers & Blogs



Secondary sources
(primarily news websites)



CoinGecko

Market cap & tokens distribution



Users
13 million**



Total Value Locked
\$ 41.5 billion

DEFI



1. PancakeSwap – 2.85M

NFT



2. AtomiAssets – 966K

GAMES



3. Alien Worlds – 908K

GAMES



4. Splinterlands – 563K

NFT



5. OpenSea – 377K

DEFI



1. Sushi - \$10.2 bln

DEFI



2. Uniswap v2 – \$7.8 bln

DEFI



3. PancakeSwap – \$6.3 bln

DEFI



4. Alcor.exchange - \$1.7 bln

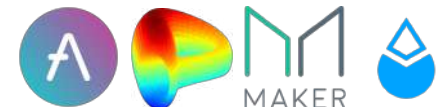
DEFI



5. AutoShark – \$1.5 bln

** AMU: Active Monthly Users (number of unique address interacting the DApp)

Low users, but high TVL





Users
13 million**

Users interacting with the top 100 DApps increased by more than **200%** in one year

DEFI



1. PancakeSwap – 2.85M

NFT



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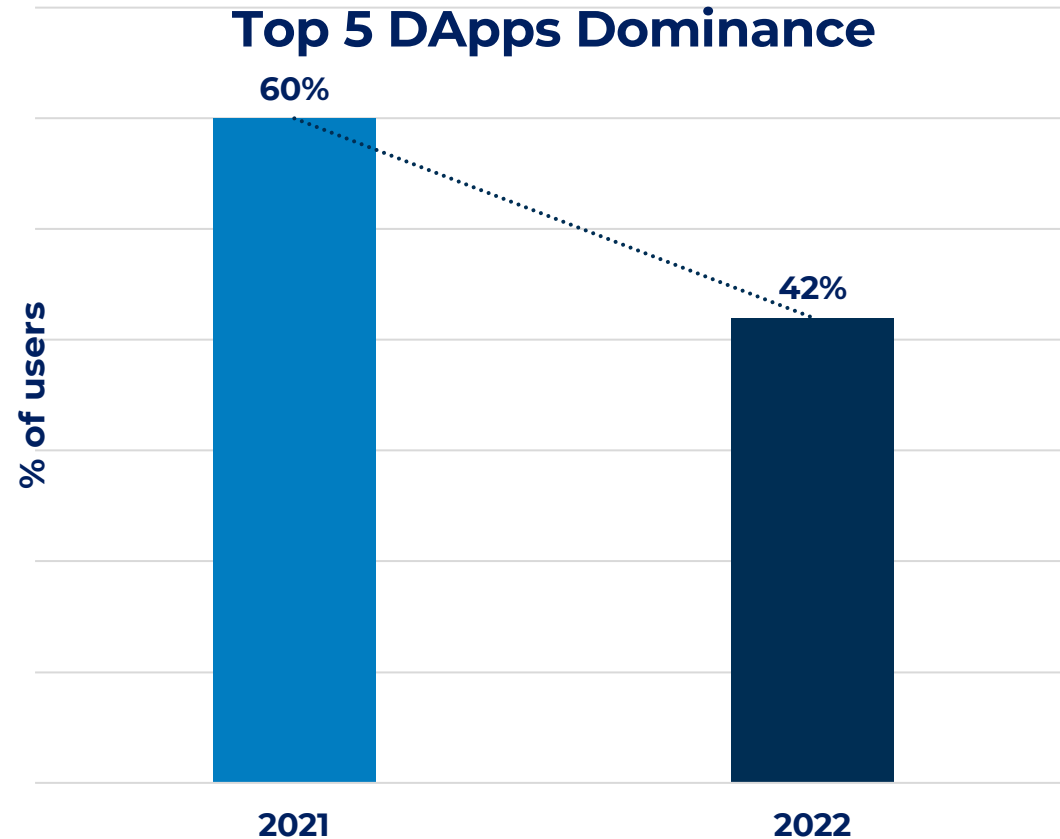
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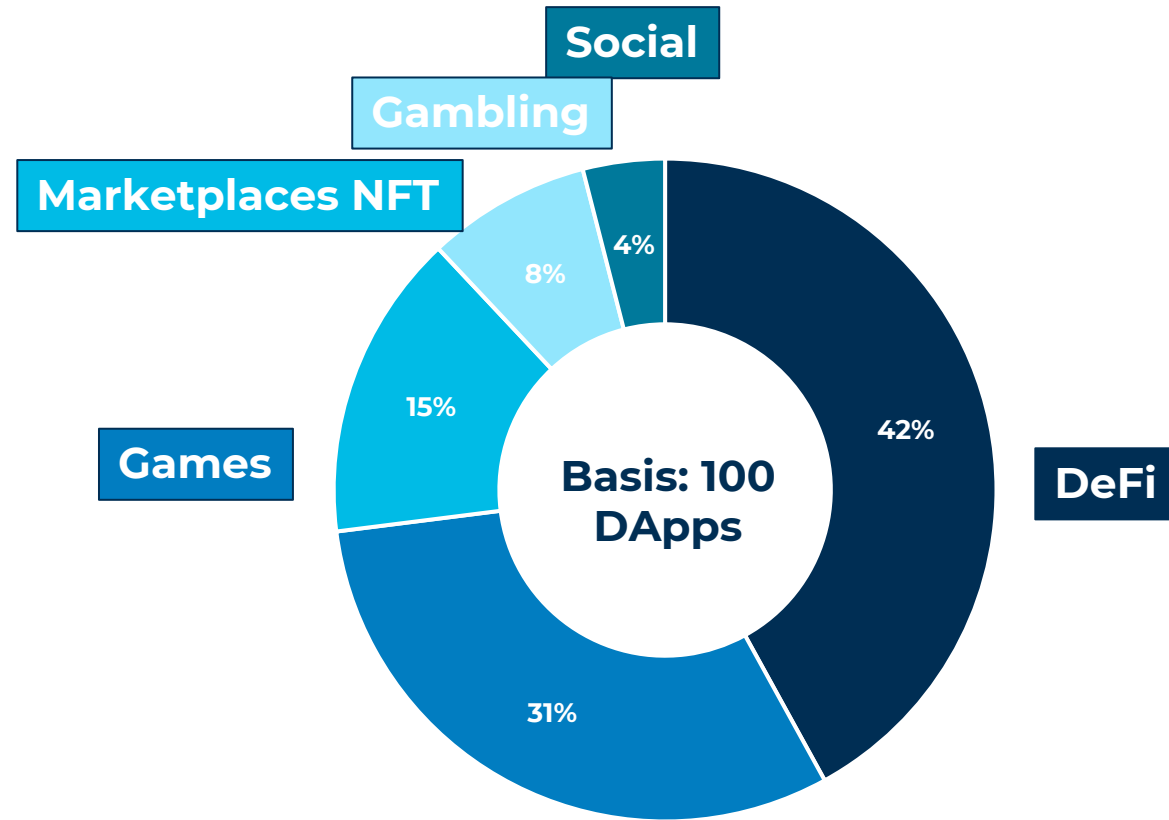
13.3 mln users

4.4 mln users

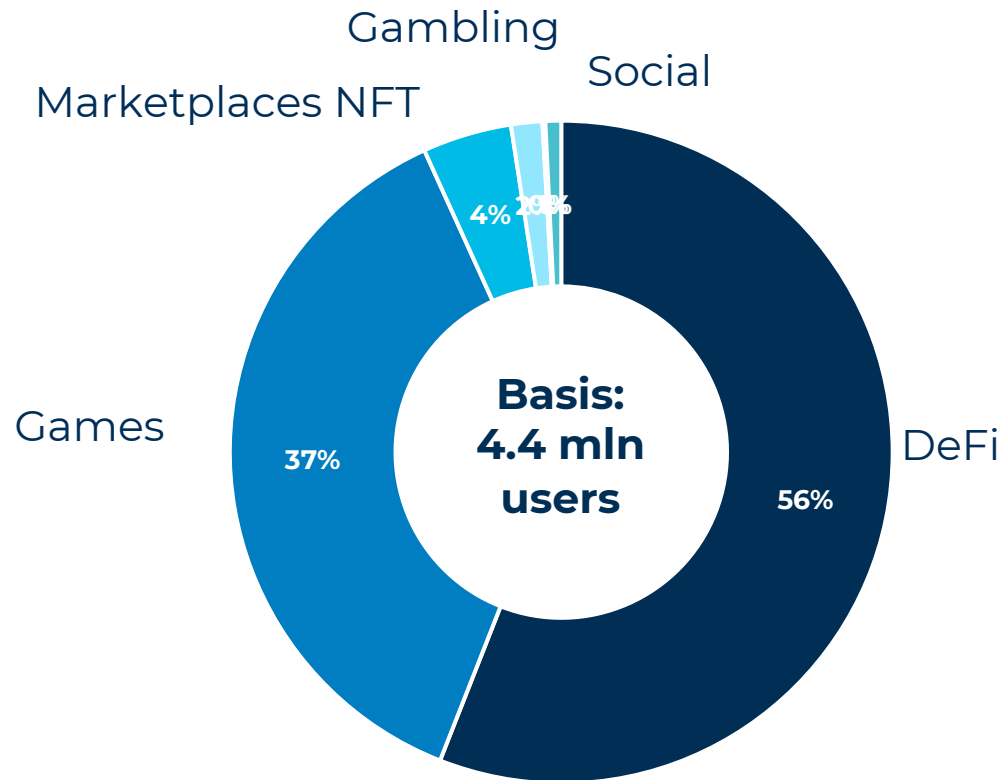
2021 users

2022 users

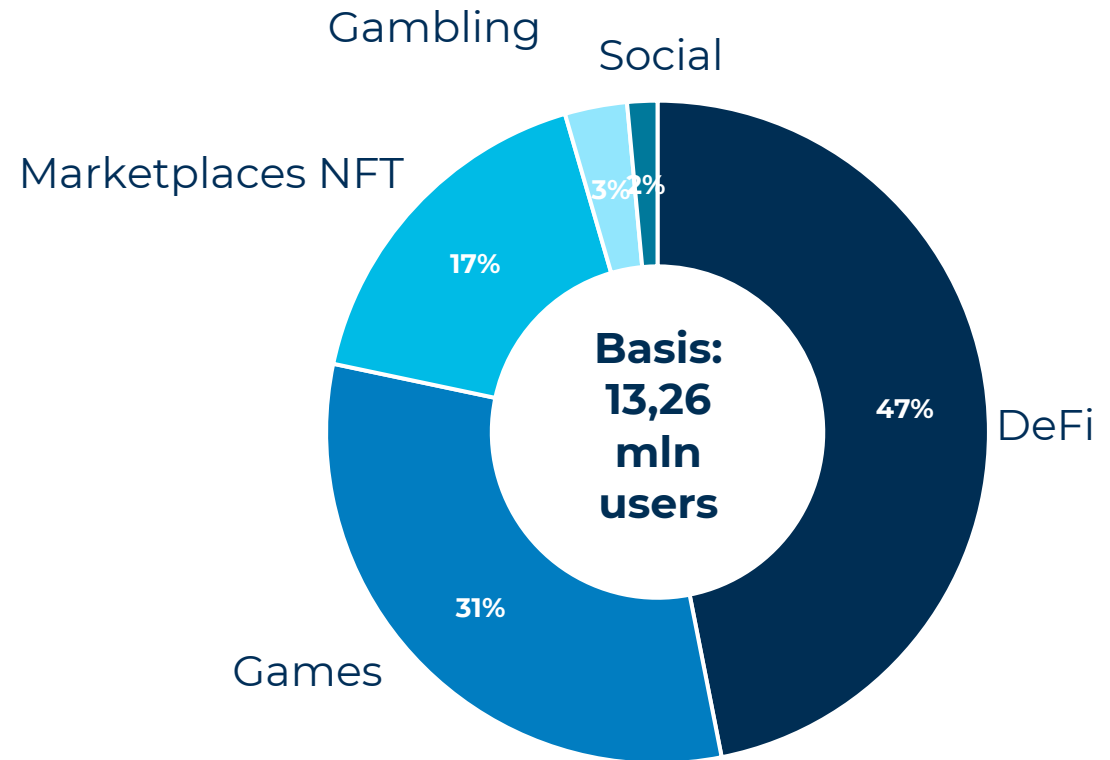




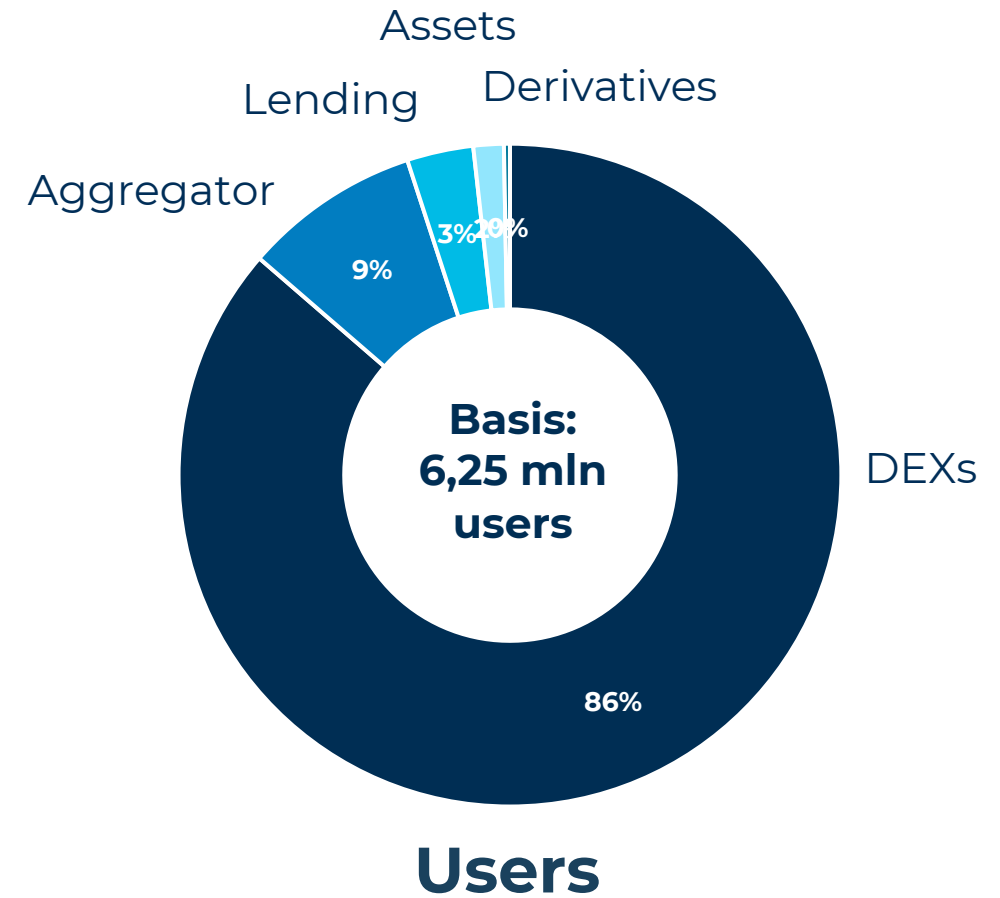
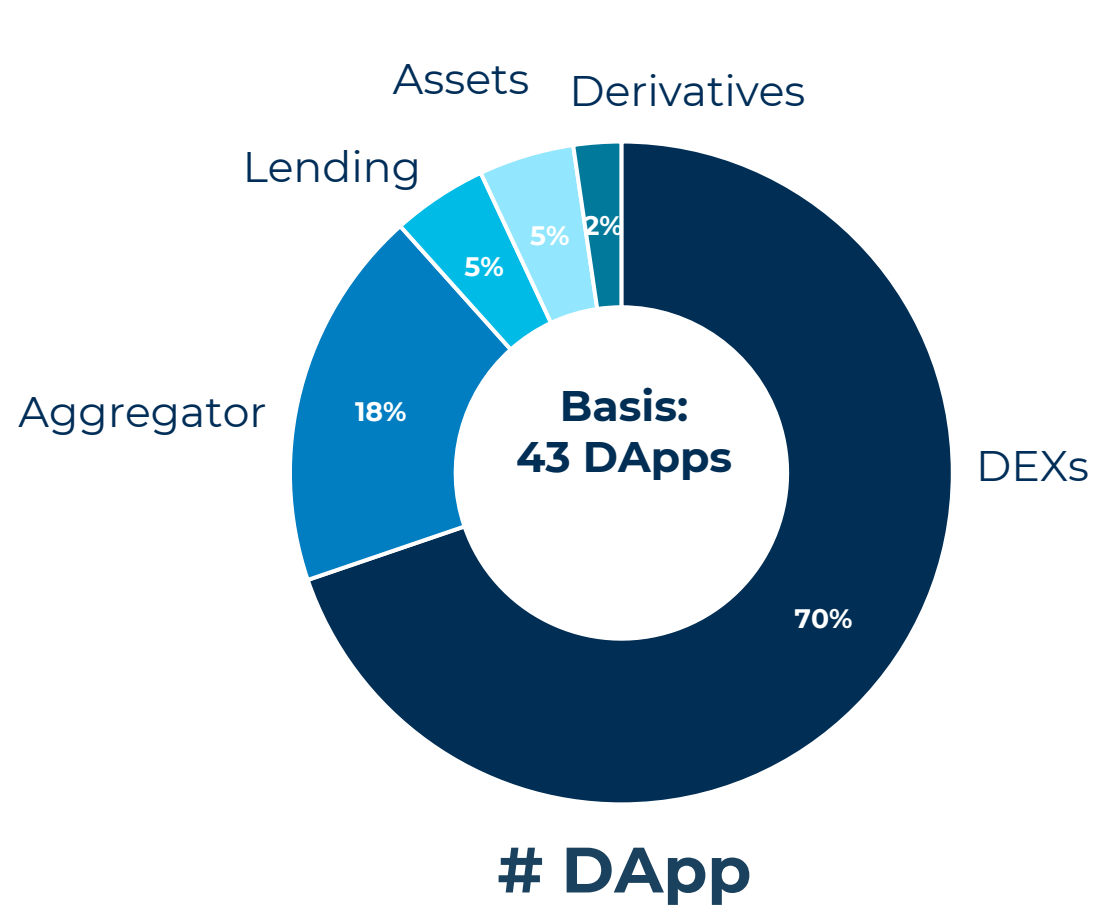
Industry



**Users
2021**



**Users
2022**



DEX: Decentralized Exchange

Aggregator: unified DEXs explorer

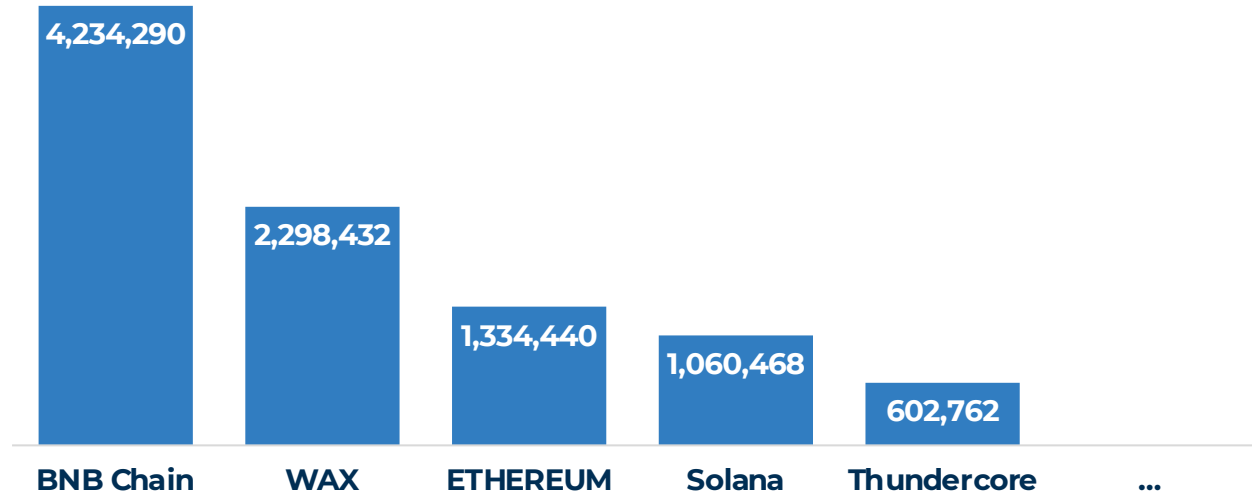
Lending: protocol for lending and borrowing,

Assets: protocols for asset management

Derivatives: tokenized derivative assets (ex:futures)

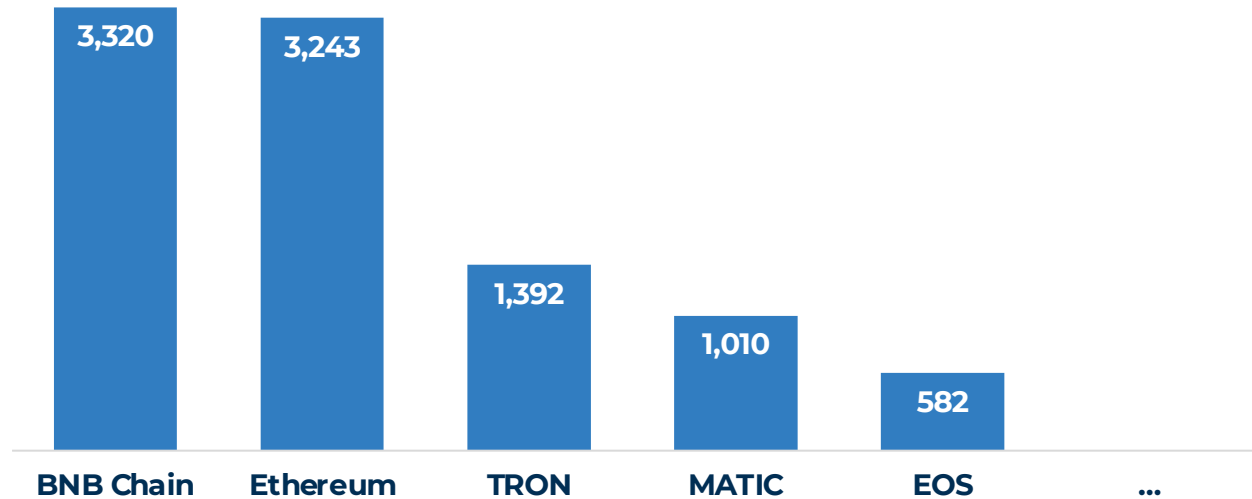
Average monthly DApp users per blockchain platform

Based on top 100 DApps



Number of DApps per blockchain platform

Total number of DApps



Source: DAppradar.com, 03/05/2022

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Composability in web3: the PoolTogether Case

Leighton Cusack
Co-founder @ PoolTogether

- | | | | |
|-------|---|-------|---|
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Tokenomics in web3

Jacopo Fracassi

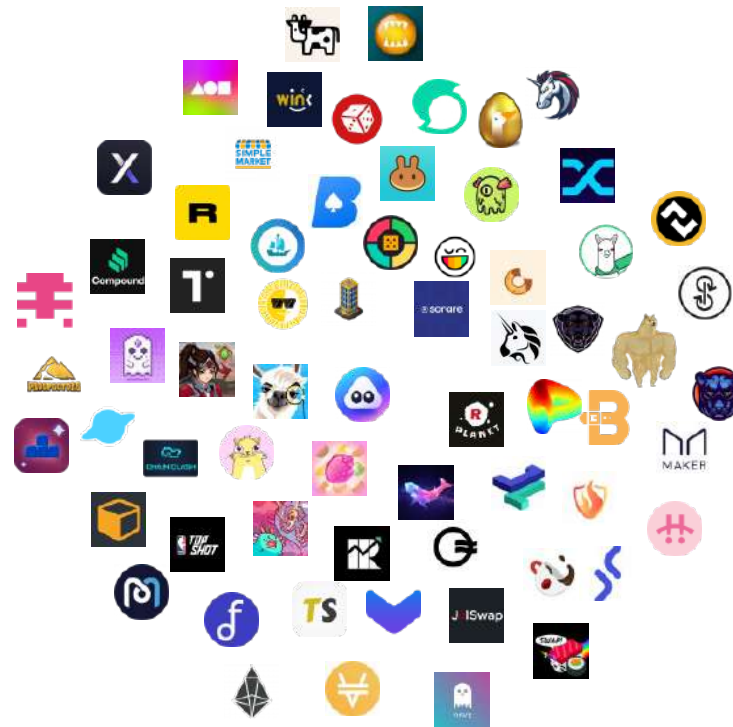
Blockchain & Distributed Ledger Observatory

There is still little research on the new architectural, business and governance models enabled by the web3

New revenue models

Decentralized Governance

Increased users' involvement



Use of token

Increased collaboration

New forms of competition



Tokenomics

**New forms of
competition**

Use of token

**New revenue
models**



Governance

**Increased
collaboration**

**Decentralized
Governance**

**Increased users'
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Tokenomics

**New forms of
competition**

Use of token

**New revenue
models**



Governance

**Increased
collaboration**

**Decentralized
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- What is tokenomics and what does it involve
- Incentive and utilities for fungible tokens
- Incentive and utilities for non-fungible tokens
- Token distribution
- Issuing strategies



Tokenomics, short for 'token economics' is an umbrella term used by cryptocurrency enthusiasts to describe how a token is used inside the project ecosystem, or how the token will follow a monetary policy as the project grows over time.



Tokenomics is the study of how cryptocurrencies work within the broader ecosystem. This includes such things like token distribution as well as how they can be used to incentivize positive behaviour in the network.

No exact definition

Combination of:



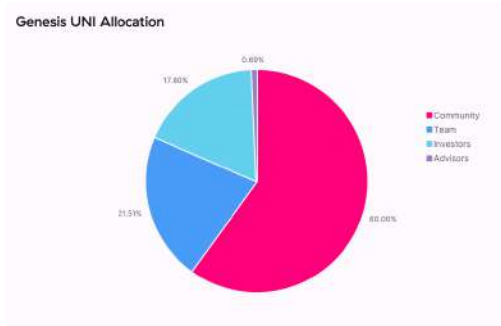
Token



Economics

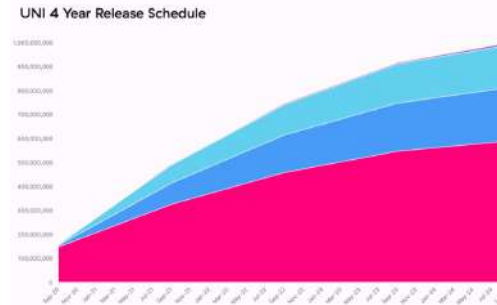
No exact definition

Includes:

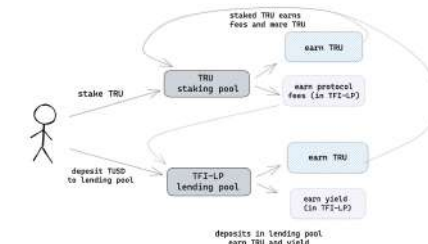


Tokens distribution on the market

Economics



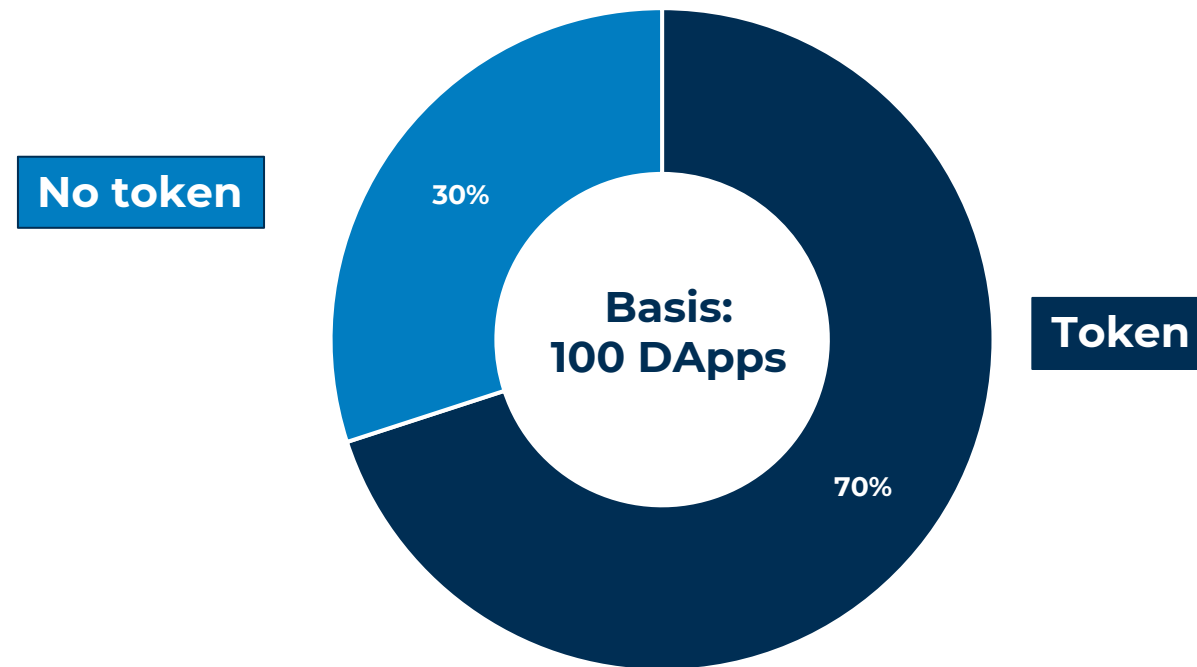
Token issuing rules



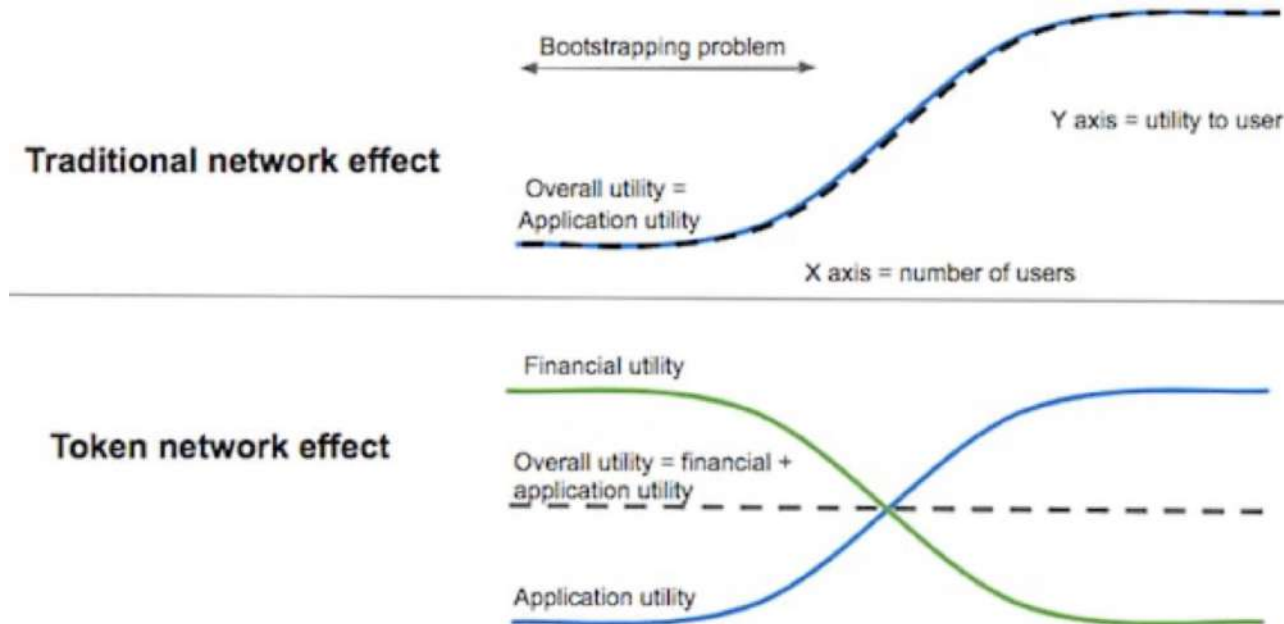
Incentives & utility

Psychology and behaviour

70% of DApps have an own token



Web3 introduces a powerful **new tool** for **bootstrapping networks: token incentives**.





Users side

Many different models.
They can be applied **stand-alone** or **combined**



Ecosystem tokens



Revenue sharing



Governance



Farming

Many different models.
They can be applied **stand-alone** or **combined**



Ecosystem tokens

Tokens that can be spent within an ecosystem. They were the first kind of utility tokens. Used by many ICOs their success depends on the products/services that the Dapp can create around them.



Revenue sharing



Governance



Farming



ApeCoin



Chainlink

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ApeCoin



Chainlink



Revenue sharing

A common incentive associated to tokens that cannot be spent within the ecosystem is to give you a reason to hold these by rewarding you. Many protocols have a fee switch system that can be activated to distributed platform fees to holders.



Governance



Farming

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Governance

Another incentive that could be considered interesting for certain users is being involved in shaping the future of a DApp. Governance is therefore another kind of utility that could make users invest in and hold a token.



Farming

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Farming

It is a different version of the Revenue sharing model, here holders are rewarded for just locking their tokens without selling them and they are rewarded with new tokens created by the protocol. It is a scheme generally used in Dapps with lower utility for their tokens.



Many different models.
They can be applied **stand-alone** or **combined**



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Governance

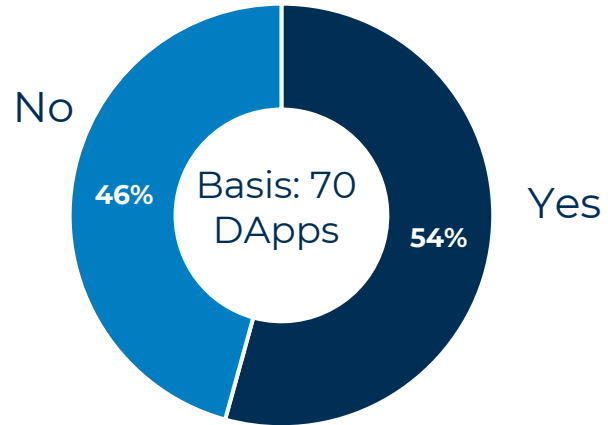
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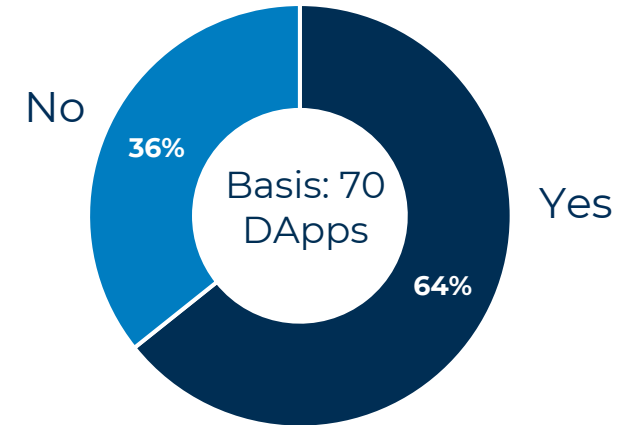
Farming



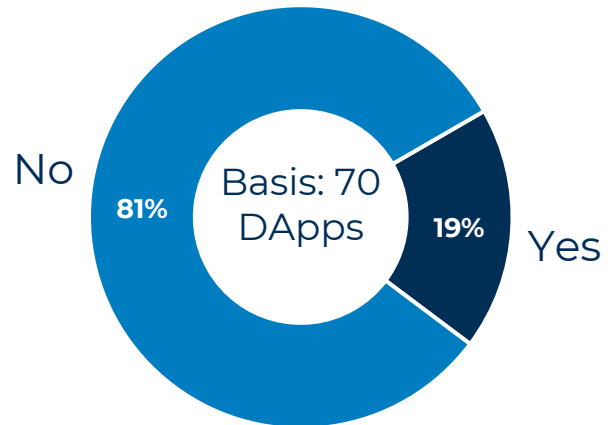
Ecosystem Token



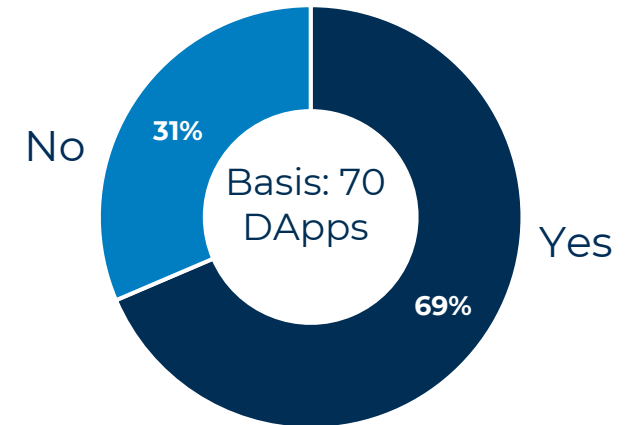
Governance Token



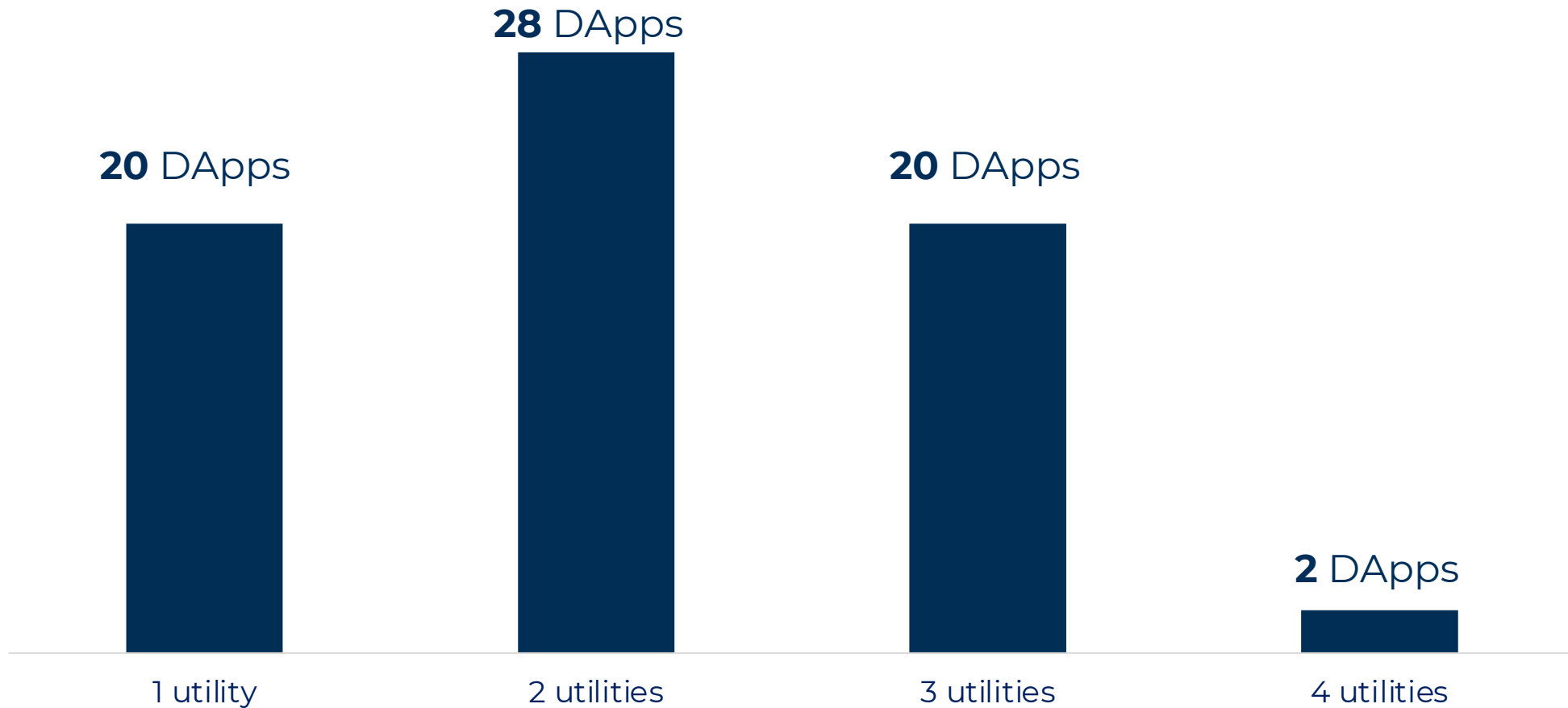
Revenue Sharing Token



Farming Token



The most common type of **cross-utility tokens** are **governance and farming** tokens



Some of these models are actually being deployed for **NFT collectibles** too.
But they are also developing **additional incentives and utility models**.



Governance

Some NFT collectibles can be used as a governance token within a DAO. In that way users can get utility by being involved in the DAO decision that usually involve how to use the collection treasury.



Farming

Some NFT collectibles can let you earn additional tokens by farming/staking them. These tokens can be a fungible one or another kind of NFT.



Coming from models already applied by Dapps for their fungible tokens

Some of these models are actually being deployed for **NFT collectibles** too.
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Farming

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Whitelisting

Owning a specific NFT can make you eligible to be whitelisted for future collections from the same artist or by some affiliated artists.



Airdrop

Owning a specific NFT can make you eligible for an airdrop. This was one of the first ways of creating additional value to NFT holders.



In-game asset

NFTs can be used as in-game assets in many different virtual worlds and games (e.g. Arcade land, The Sandbox, Decentraland)





Users side

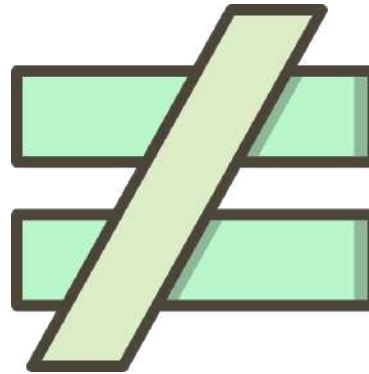


Users side



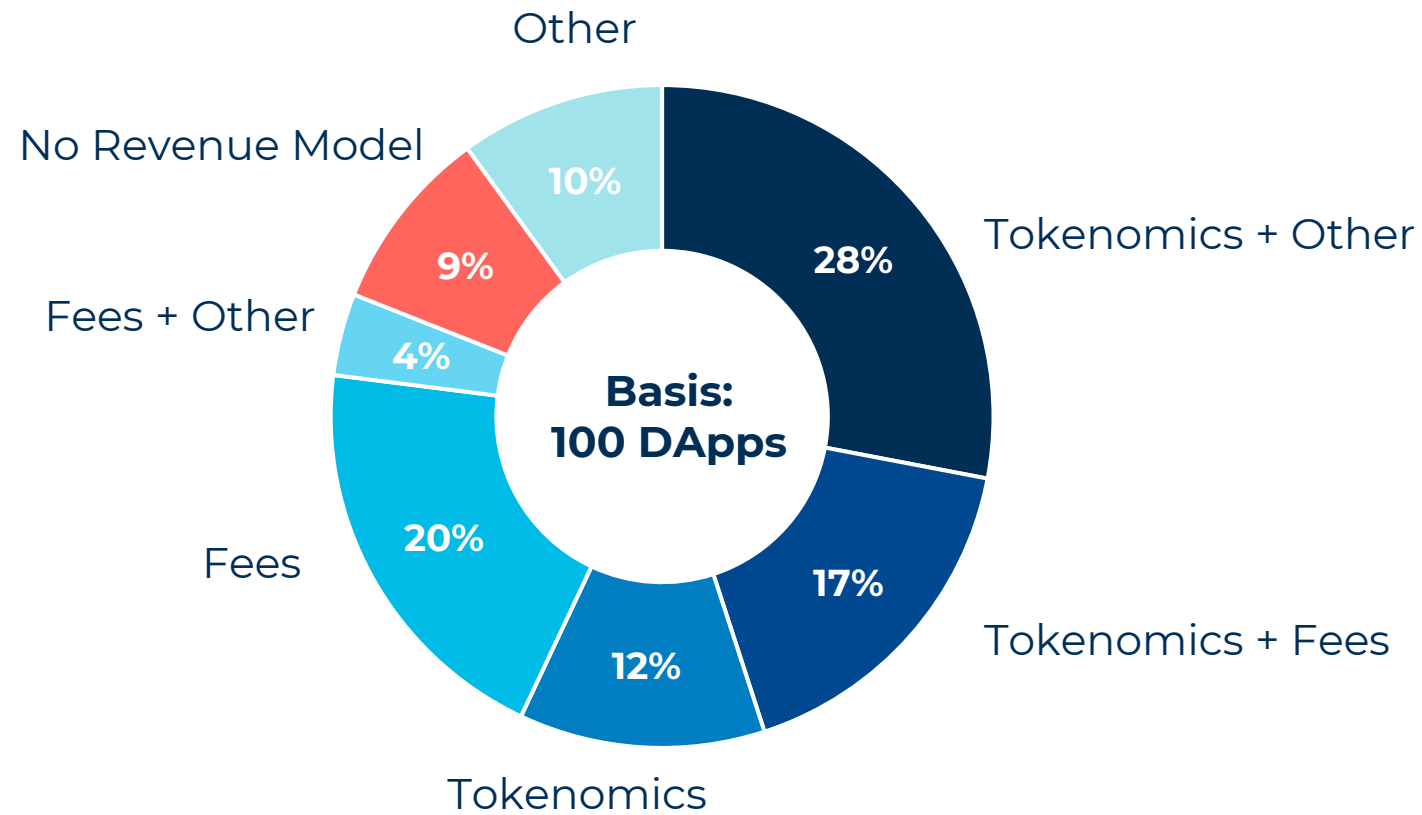
Revenue side

Every token has a specific tokenomics

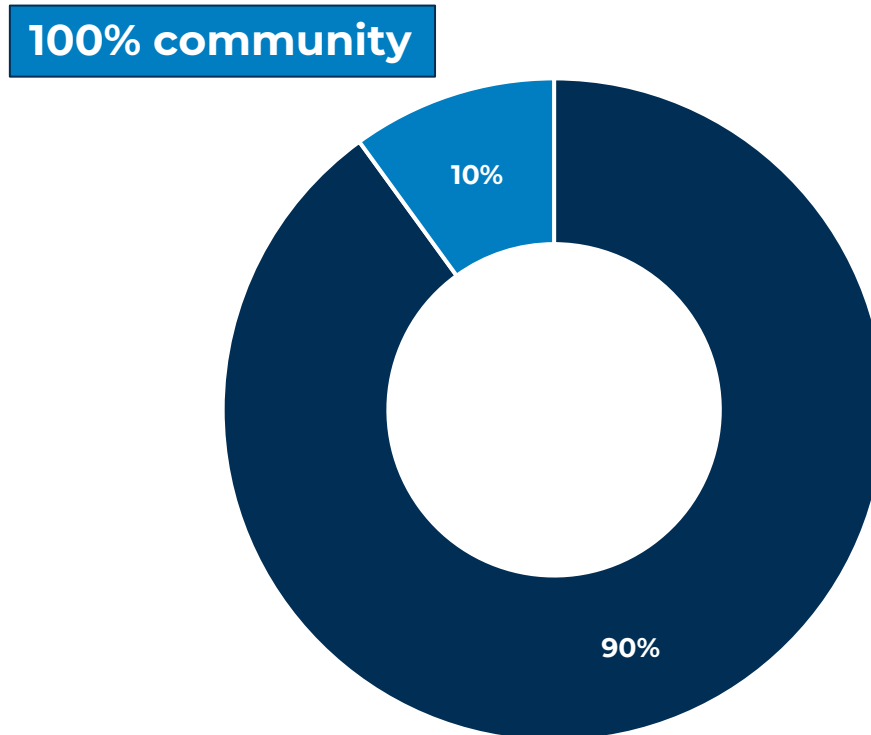


Not every Dapp with a token uses tokenomics as a revenue model

Tokenomics is the most common revenue model for Dapp owners

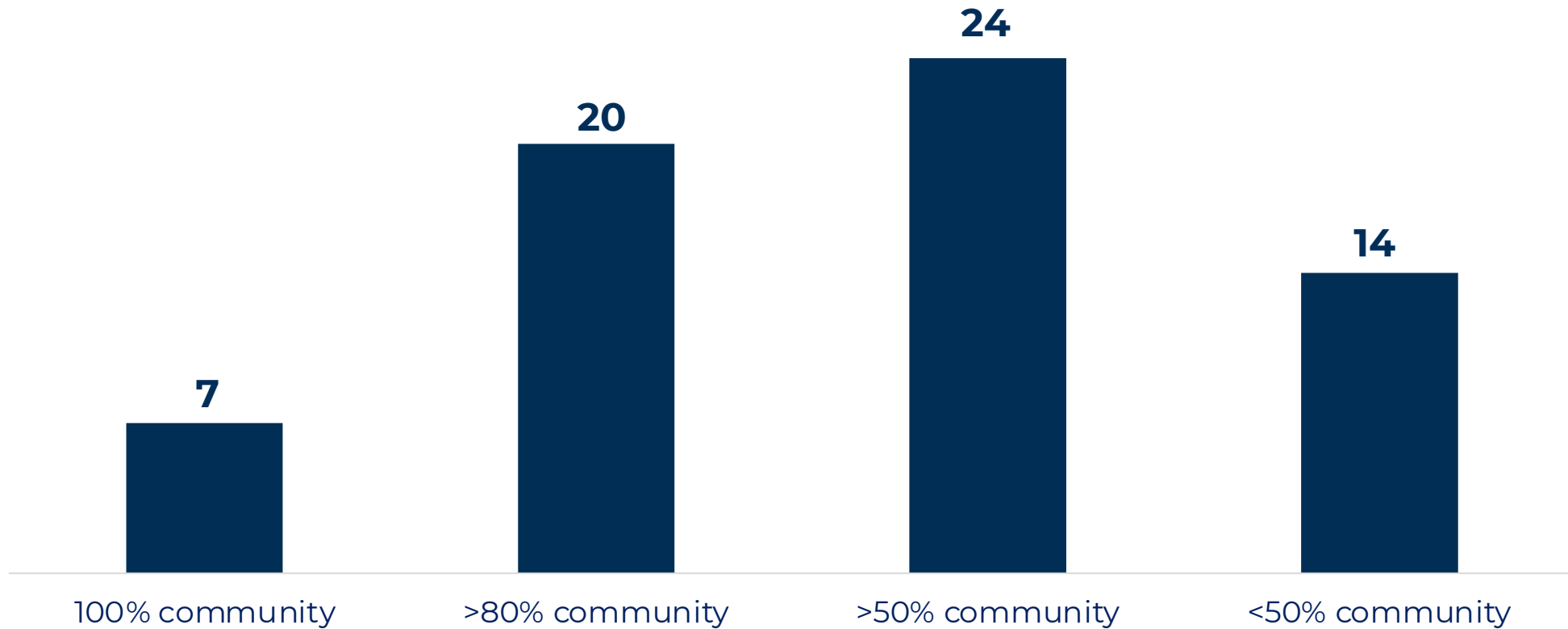


Distribution patterns for Dapps with a token

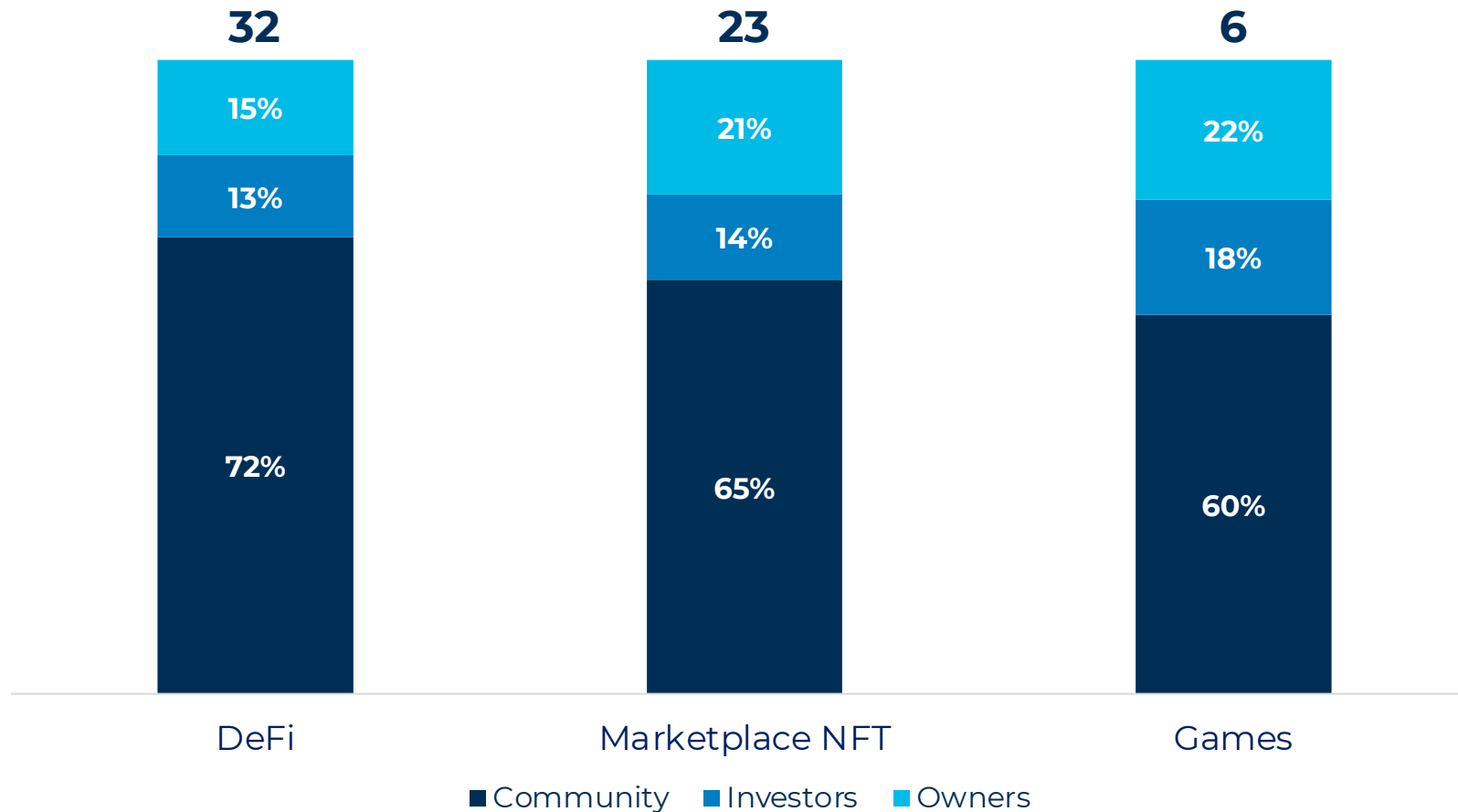


Basis: 70 DApps with a token

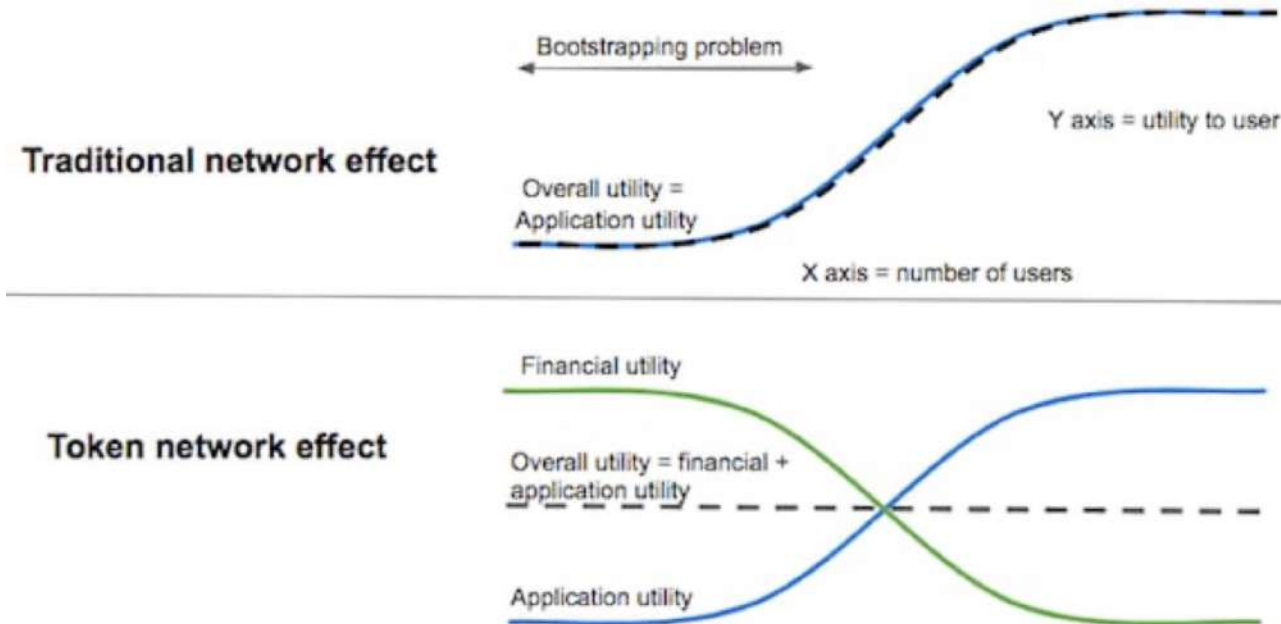
Distribution pattern for DApps with a token and a declared initial distribution



Average token distribution for DApps using tokenomics by category



Web3 introduces a powerful **new tool** for **bootstrapping networks: token incentives**.



- **Initial Coin Offerings (ICOs)**
- Initial DEX Offering (IDOs)
- Retroactive airdrop
- Usage rewards
-

Description:

this was the first distribution mechanism adopted to get DApp tokens into the wild. Its name mimics IPOs, but it used to be unregulated and it was done through blockchain platforms (e.g. Ethereum).

Years: 2014-2018

Pricing: deterministic

Alternatives: Private sales, donations

Liquidity: low

Risks:

- Nowadays could be assimilated to securities issuing
- Low liquidity after token issuing (only selected exchanges or no liquidity at all)
- Usually there wasn't a market for the token issuing
- Rush sales

- Initial Coin Offerings (ICOs)
- **Initial DEX Offering (IDOs)**
- Retroactive airdrops
- Usage rewards
-

Description:

This issuing strategy is used by many Dapps to create from day 1 a market for their token. A Dapp can use a decentralized exchange like Sushi or Uniswap and add their token with their paired token so people can start trading on it.

Years: 2019-ongoing

Pricing: target price

Alternatives: IEO, Liquidity bootstrapping pools

Liquidity: high

Risks:

- Careful target price definition is needed
- High fluctuation in value after the IDO
- Higher initial capital requirements to provide liquidity
- Could also require an incentivized liquidity

- Initial Coin Offerings (ICOs)
- Initial DEX Offering (IDOs)
- **Retroactive airdrops**
- Usage rewards
-

Description:

This issuing strategy rewards early supporters of a Dapp. This can be based on historic app usage, holding an NFT, being part of another project, or even just being early in the Discord. Usually, this issuing strategy is not applied as a standalone.

Years: 2020-ongoing

Pricing: none

Alternatives: -

Liquidity: depends on additional issuings

Risks:

- Tokens dump after the airdrop (incentives to avoid that could be applied)
- Airdrop farmers with multiple wallets
- Initial vesting should be defined for investors and developers

- Initial Coin Offerings (ICOs)
- Initial DEX Offering (IDOs)
- Retroactive airdrops
- **Usage rewards**
-

Description:

Rewarding people for using your app is widely considered a key element of web3. The reward could be distributed for very different activities, such as: providing liquidity, trading through a platform, playing, walking (e.g., STEPN)...etc.

Years: 2020-ongoing

Pricing: none

Alternatives: liquidity
bonding

Liquidity: depends on
additional issuings

Risks:

- How to distribute the rewards and how much of them should be defined carefully
- Its effect is usually temporary
- Rewards dumping could be an issue

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Governance models in web3 projects

Davide Ghezzi

Blockchain & Distributed Ledger Observatory



The growth of a web3 project often involves the creation of a **community**

Not only are the number of DApps users growing, but also those who **actively contribute** to the development of these projects



Building **decentralised governance** means **creating a collaborative development protocol**, actively involving the user base through a **defined set of rules**

Blockchain technology enables the **governance** of a project to be managed in an unprecedented way. **Governance becomes an innovative element** for DApps and web3 projects development

Users not only collaborate, but can **actively participate** in key **development decisions**

- **How** do users get involved?
- **How much** are users involved?
- **What** kind of decisions are made?
- What is the **purpose**?

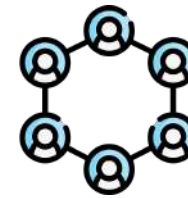
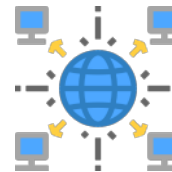
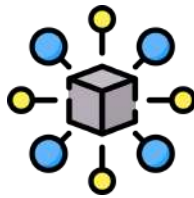
What is the **purpose** of decentralising governance?



- Attract new **users**
- **Open innovation**
- Enabling **tokenomics** models
- **Overcoming the limitations** of current systems
- To **build a community** around a **common project**, driven by a **common purpose**

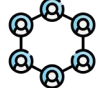
Decentralised governance is not a binary property, as decentralisation can take place at different levels

There is a **spectrum** of possible decentralisation levels

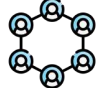


**Centralized
governance**

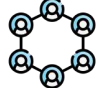
**Decentralized
governance**

		 Completely centralized	Potential spectrum	 Completely decentralized
 Governance	- Who decides what can be altered by governance token holders? - Who can vote protocol changes? - Are all governance tokens freely traded?	Only operators can change any aspects of the system		All aspects can be altered, any token holder can propose changes
 Custody	- Who is in charge of guarding the assets - Does the user retain control over funds at all times? - Who controls the admin keys?	A single entity retains full control of assets		Users have always full control of assets
 Protocol modification	- Who can make changes to the protocol? - Once a smart contract is deployed, can the code be changed by a party unilaterally?	Operators alone can modify all protocol parameters		All protocol changes are automatically enforced by smart contracts
 Verifiable security	- Is the code deployed on-chain publicly accessible? - Has there been an audit of the code? - Has an audit report been made public?	Code is not transparent and unaudited		All aspects can be altered, any token holder can propose changes

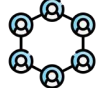


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**Building consensus
on a code change
decision**

**Enforcement of
the decision**

Building consensus on a code change decision



Enforcement of the decision

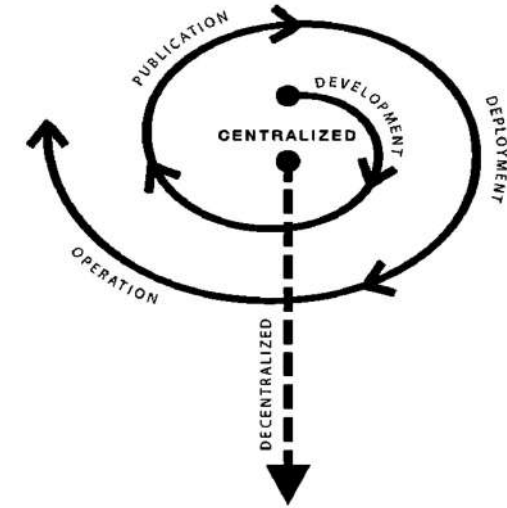




How the decentralisation process can unfold: a common pattern

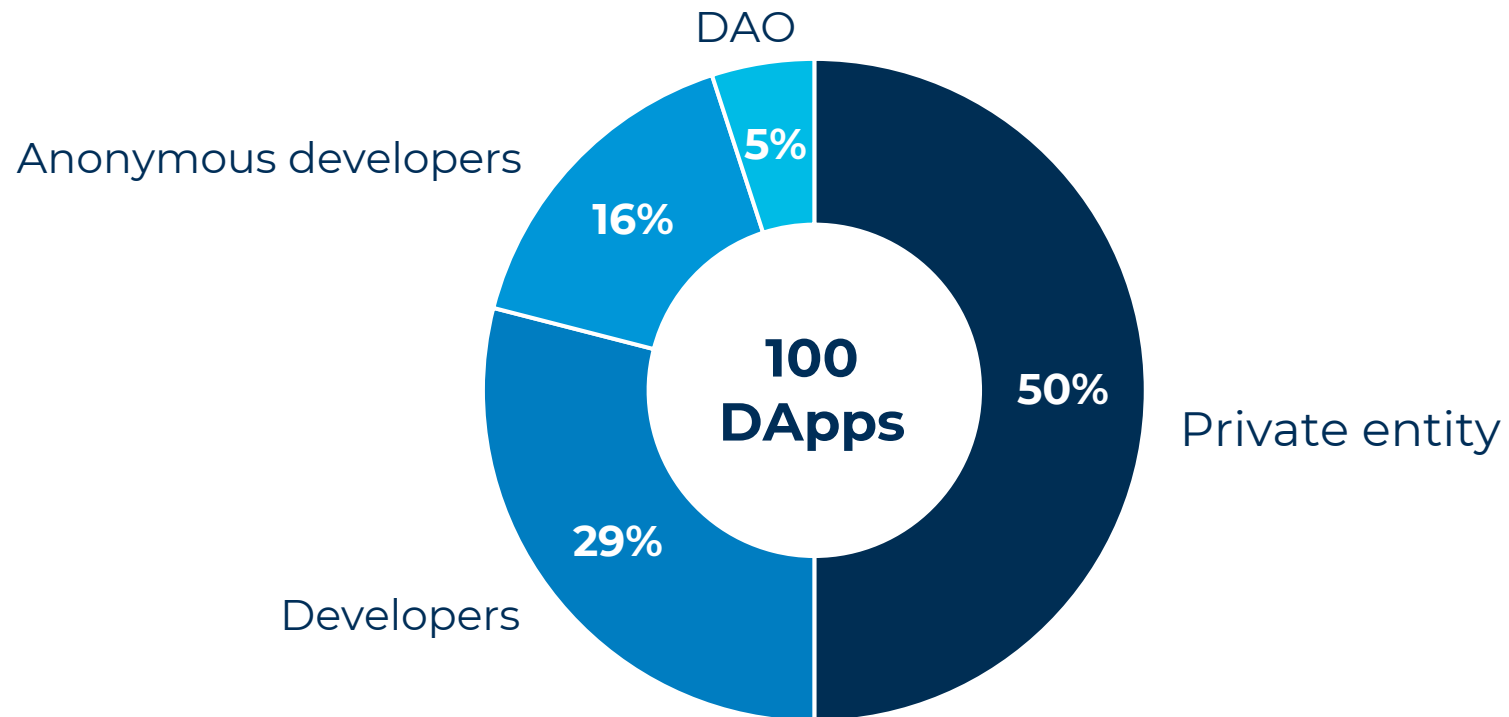
Dapps do not usually set up decentralised governance from day one, but rather build a **community** and **involve users progressively**

Some DApps eventually arrive to be **fully controlled** by a **DAO**, but **this is not always the goal**



Decentralisation of governance is not straightforward.
It is generally an **iterative process**.

Decentralized application **doesn't mean** decentralized governance



Decentralisation of governance brings several **risks**



- Loss of **control**
- **Failures** in voting processes
- **Speculation** on governance tokens
- **Plutocracy**
- ...



Blockchain can bring decentralisation at an **infrastructural** level



DApps can bring decentralisation at an **application** level

Web3 applications will be shaped by **deep user involvement** and **ownership in the project governance**

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Progressive decentralization and future of DAOs

Sebastiano Cataudo, Polygon DAO Core Dev
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Growth @ Hop Protocol

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Giacomo Vella, <i>Blockchain & Distributed Ledger Observatory</i> | 15:50 | Tokenomics and interoperability: the Hop Protocol case
Emmanuel Coen, <i>Growth @ Hop Protocol</i> |
| 14:40 | Composability in web3: the PoolTogether case
Leighton Cusack, <i>Co-founder @ PoolTogether</i> | 16:05 | Furucombo's composability approach
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| 15:10 | Governance models in web3 projects
Davide Ghezzi, <i>Blockchain & Distributed Ledger Observatory</i> | 16:30 | TECH DEMO Open sourcedness on steroids: lessons from the DApp and web3 ecosystem
Francesco Bruschi, <i>Blockchain & Distributed Ledger Observatory</i>
Tommaso Paulon, <i>Blockchain & Distributed Ledger Observatory</i> |
| | | 17:00 | End of the day |



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Furucolombo's composability approach

Blake Ho

COO @ Furucombo

- | | | | |
|-------|---|-------|---|
| 14:00 | Opening session | 15:20 | Progressive decentralization and future of DAOs
Sebastiano Cataudo, <i>Polygon DAO Core Dev</i>
Marco Moshi Grendel, <i>DAO Lead, Polygon</i> |
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Discussion and Q&A

Composability

Tokenomics

Decentralised governance

- | | | | |
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TECH DEMO Open sourcedness on steroids: lessons from the DApp and web3 ecosystem

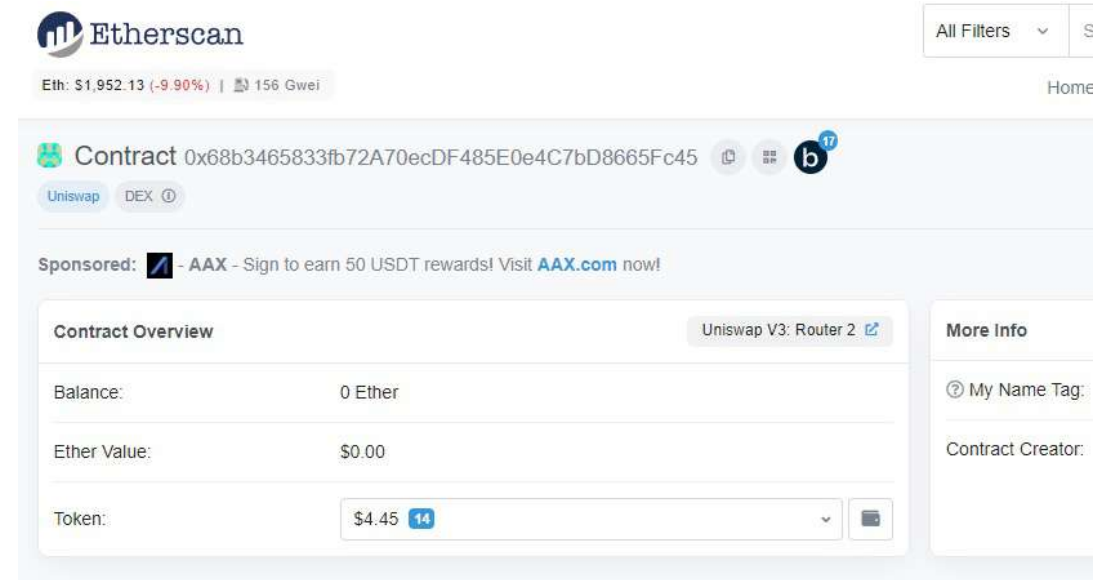
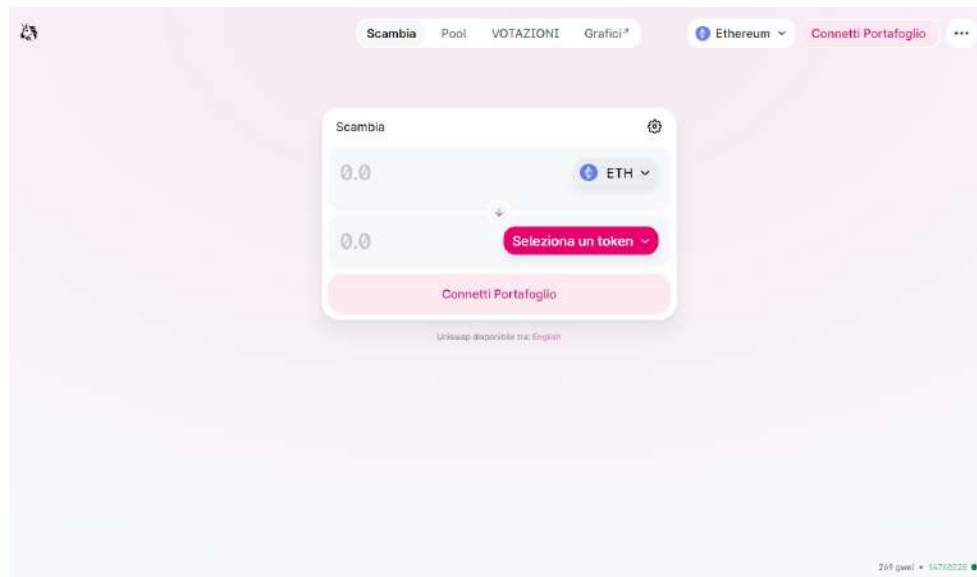
Francesco Bruschi

Tommaso Paulon

Blockchain & Distributed Ledger Observatory

A **Dapp** consists of two main components: **Smart Contracts** and **Front-end Application**

- ❖ Some computation can be carried out by a back-end, but its presence **should not be essential** for the proper functioning of the application



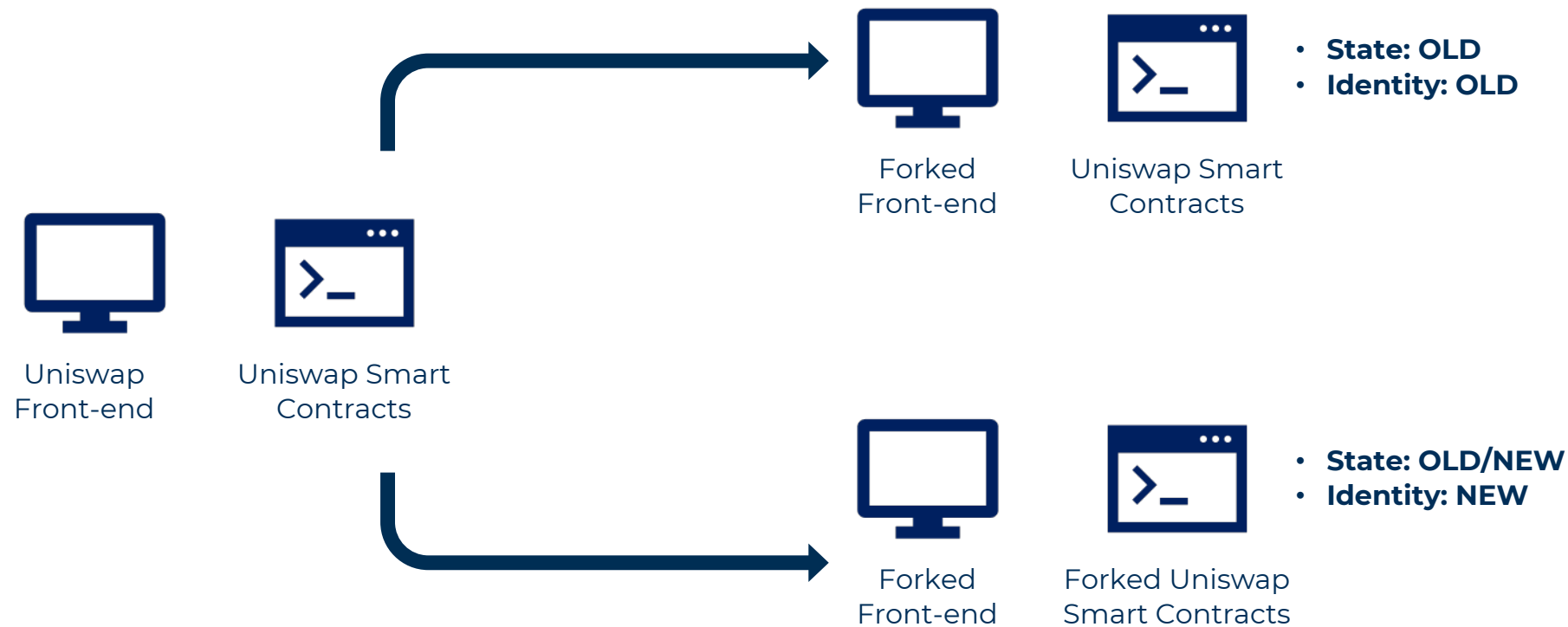
Let's imagine a decentralized version of Twitter:

- on-chain state
- decentralized storage (es. IPFS)

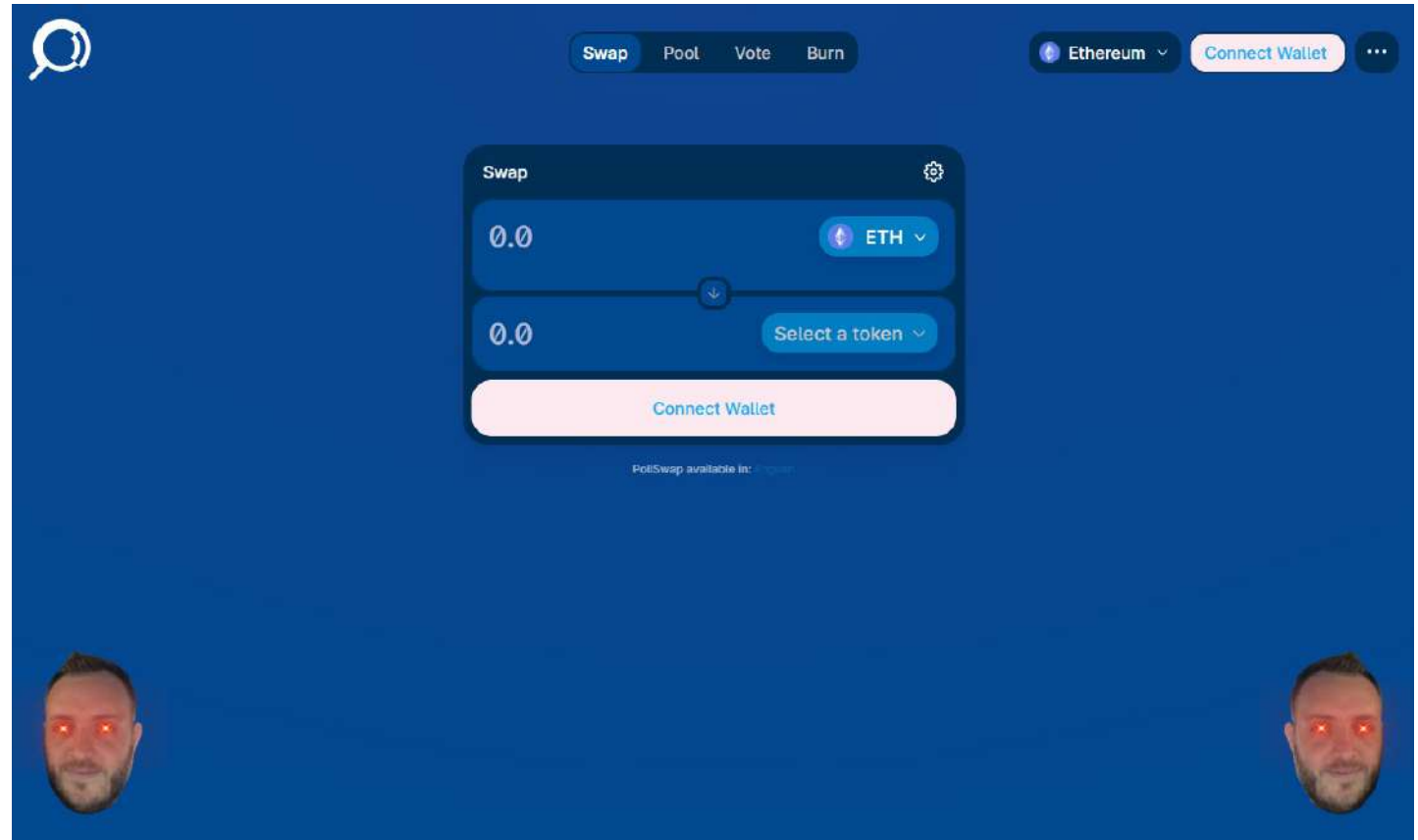
→ anyone could implement a new version with additional features (es. 'Edit' button)

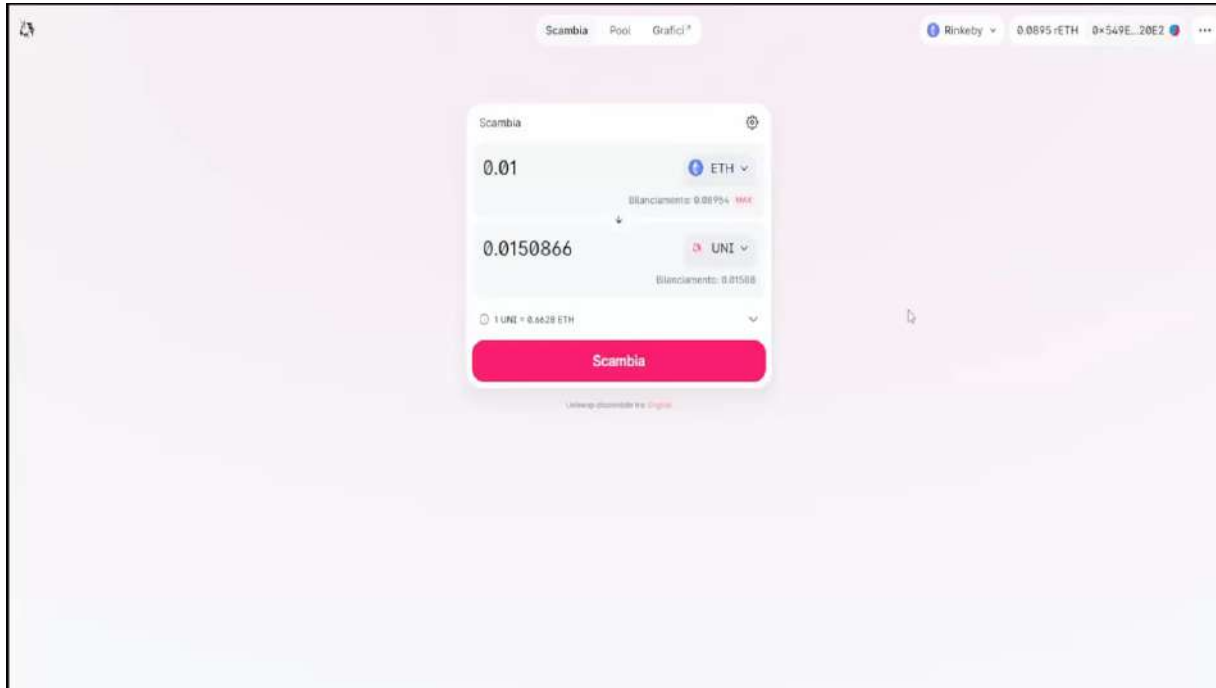


In **Uniswap** both the front-end and the smart contracts are open source thus can be forked
Forking the contracts still allows to inherit Uniswap's **state**, but not its **identity**

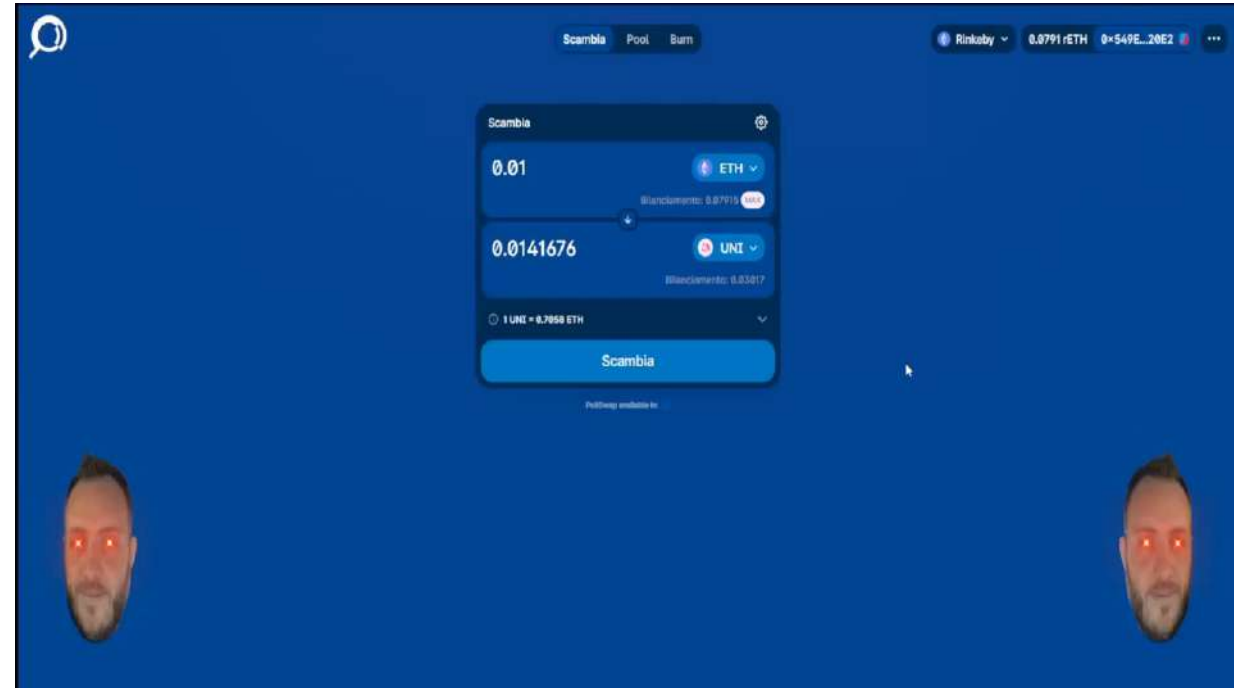


It is possible to build
a **customized web
interface** to interact
with Uniswap's
Smart Contracts,
**without having to
ask for permission**





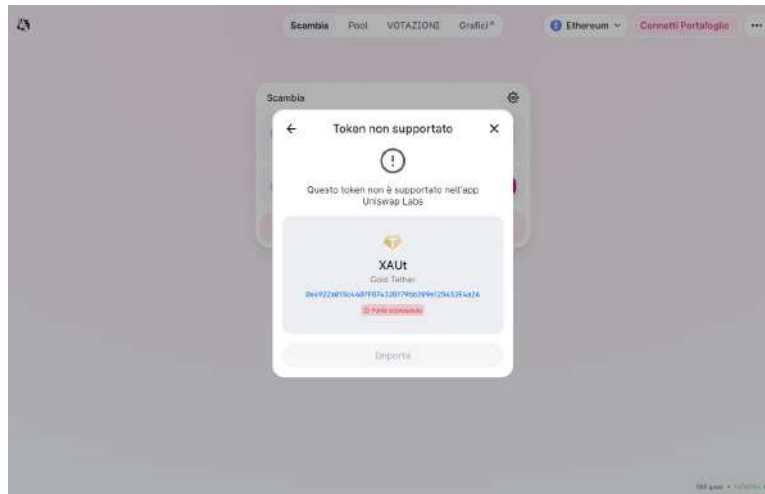
Uniswap Original Front-end



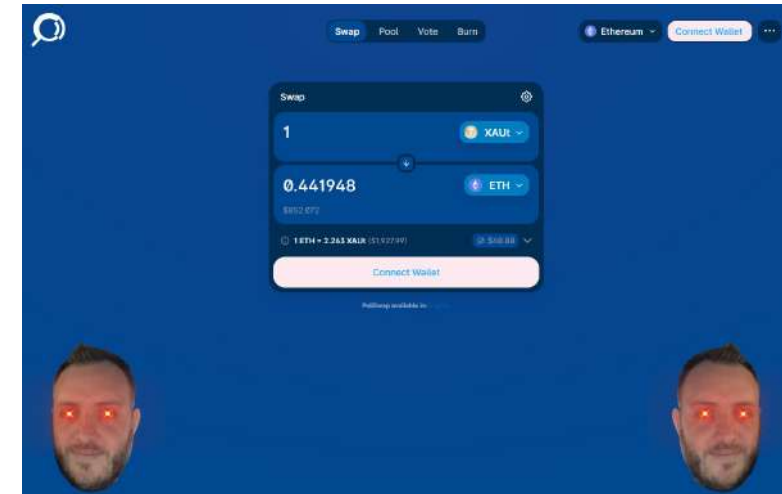
Uniswap Customized Front-end

Uniswap rimuove 100 token dalla propria interfaccia, si ritiene per timori di natura normativa

Token sintetici e azioni tokenizzate sono entrati nella lista nera di Uniswap Labs: la community



Uniswap Original Front-end



Uniswap Customized Front-end



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 13.05.22

 Blockchain & Distributed Ledger Observatory

DApp & web3: new models and opportunities